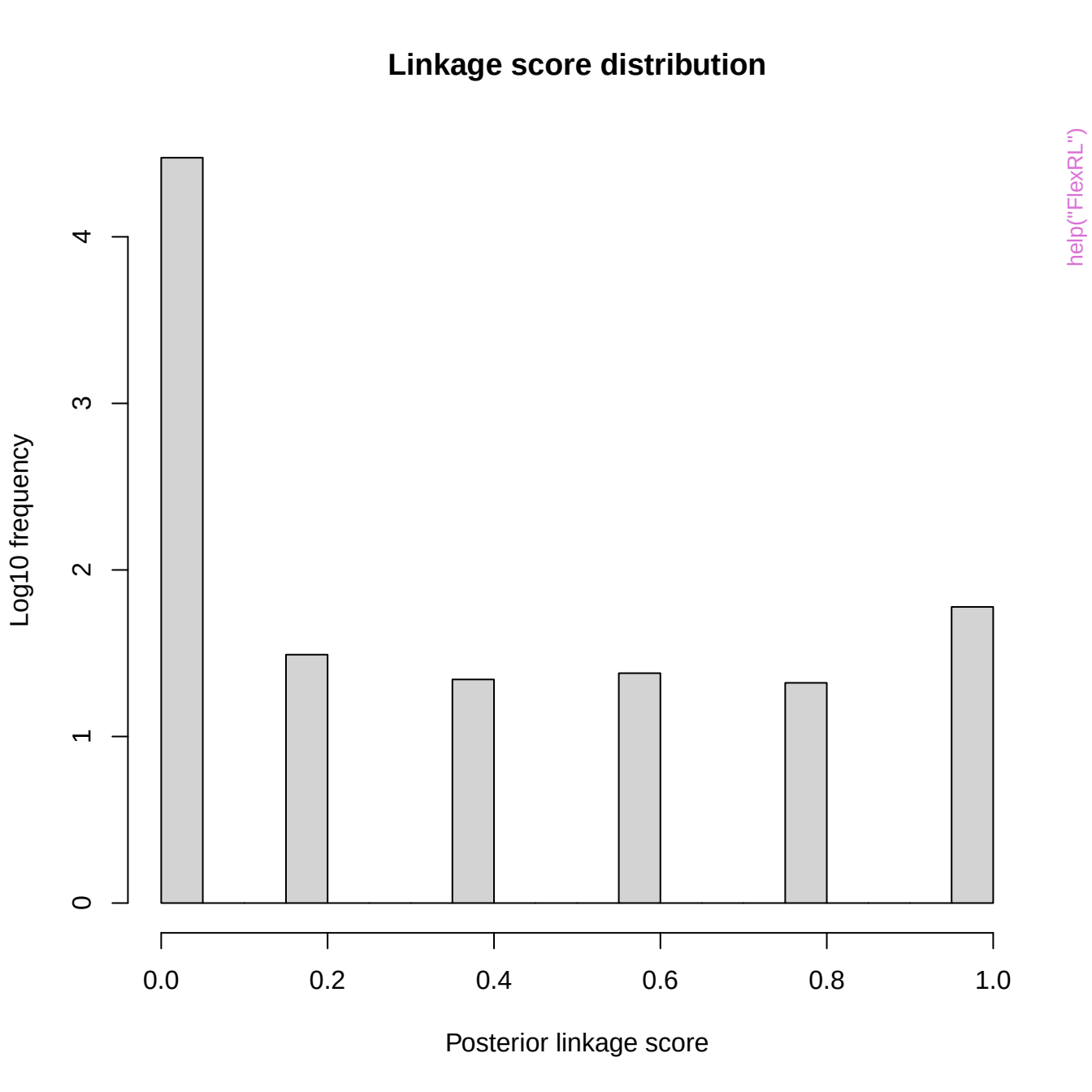
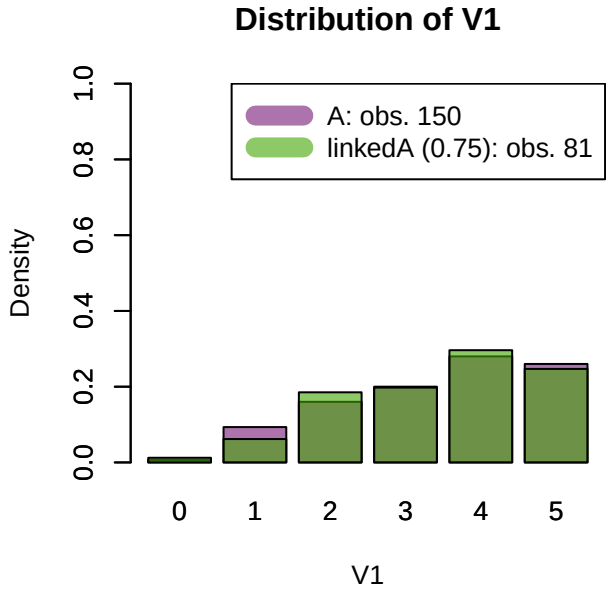
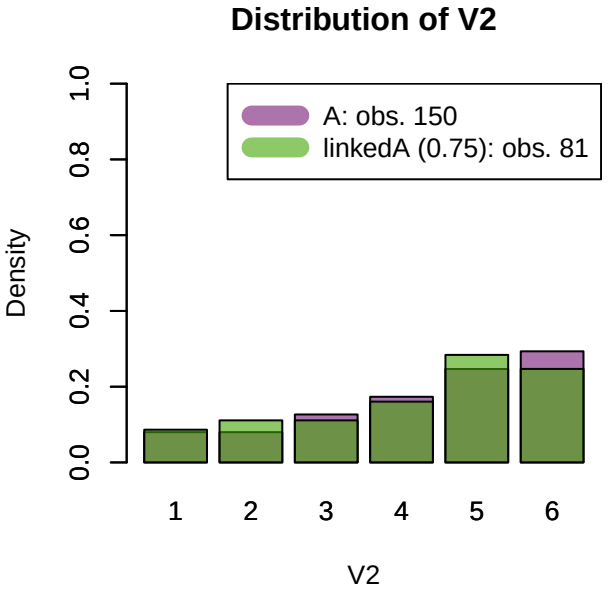
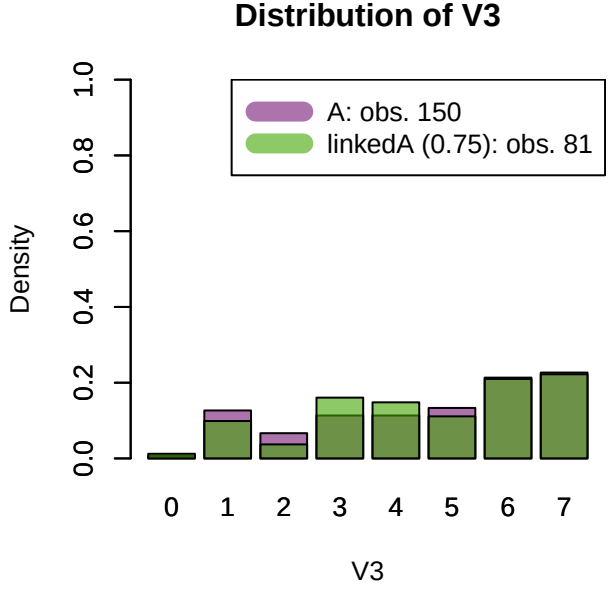
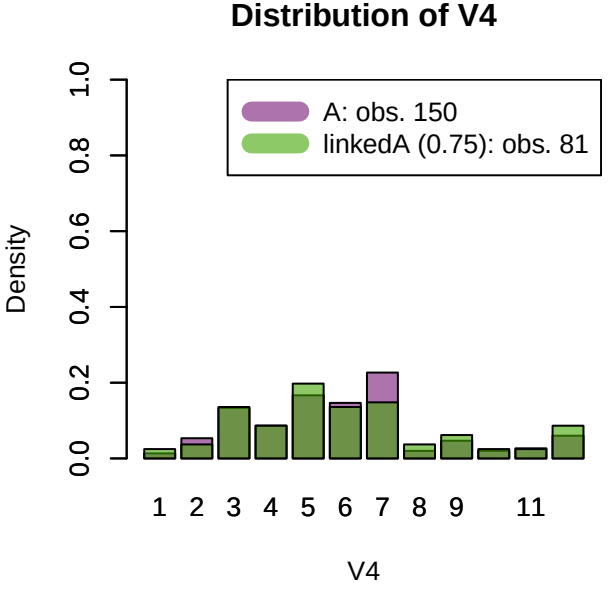

Linkage score distribution
help("FlexRL")
Log10 frequency
0
1
2
3
4
0.0
0.2
0.4
0.6
0.8
1.0
Posterior linkage score

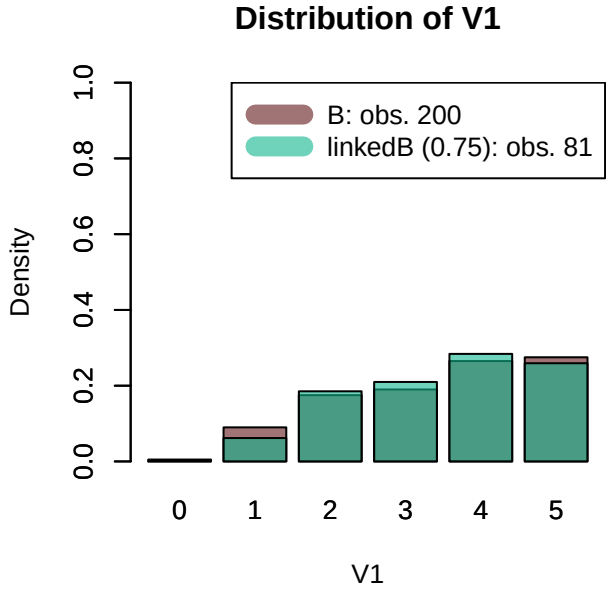

Distribution of V1
A: obs. 150
linkedA (0.75): obs. 81
Density
0.0
0.2
0.4
0.6
0.8
1.0
0
1
2
3
4
5
V1

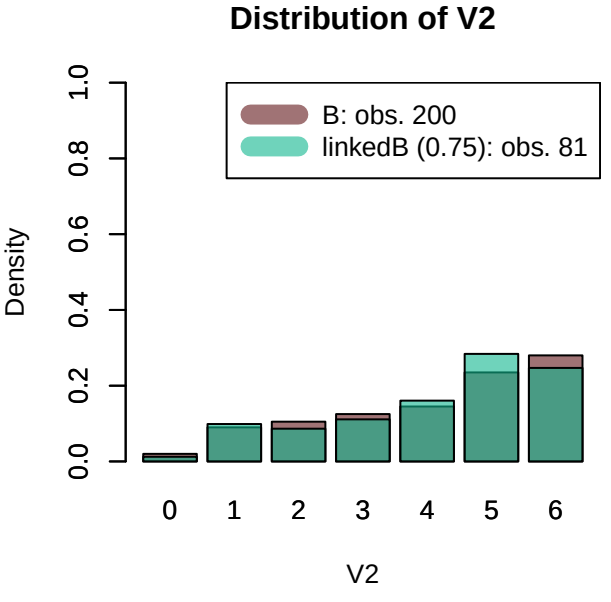

Distribution of V2
A: obs. 150
linkedA (0.75): obs. 81
Density
0.0
0.2
0.4
0.6
0.8
1.0
1
2
3
4
5
6
V2

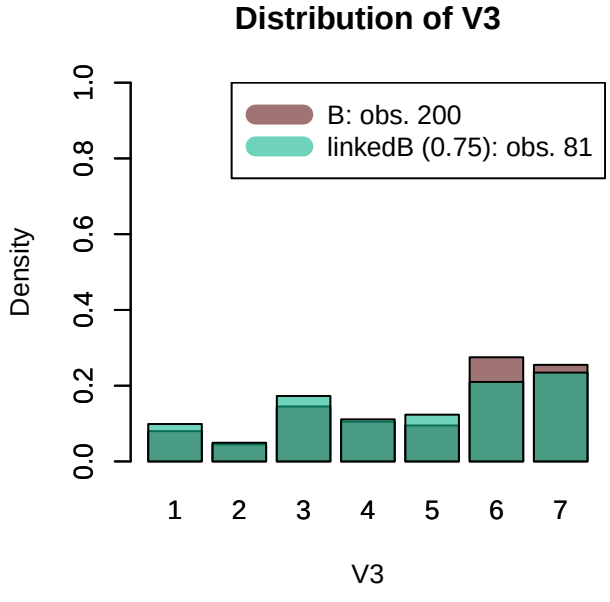
help("FlexRL")

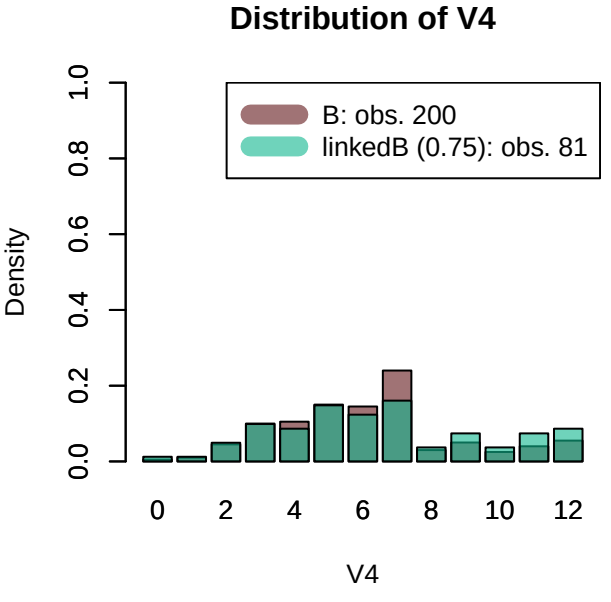

Distribution of V3
A: obs. 150
linkedA (0.75): obs. 81
Density
0.0
0.2
0.4
0.6
0.8
1.0
0
1
2
3
4
5
6
7
V3


Distribution of V4
A: obs. 150
linkedA (0.75): obs. 81
Density
0.0
0.2
0.4
0.6
0.8
1.0
1
2
3
4
5
6
7
8
9
11
V4

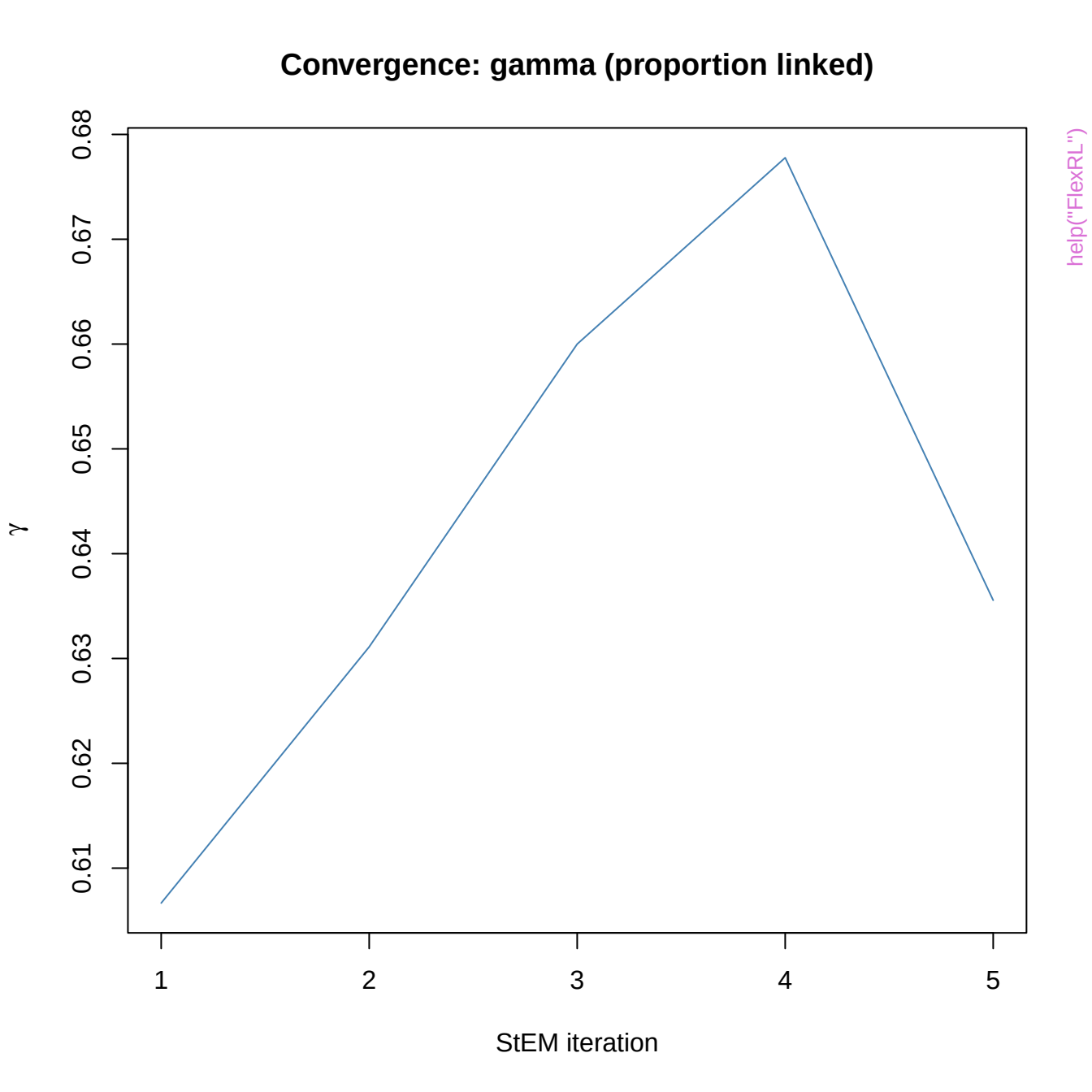

Distribution of V1
B: obs. 200
linkedB (0.75): obs. 81
Density
0.0
0.2
0.4
0.6
0.8
1.0
0
1
2
3
4
5
V1

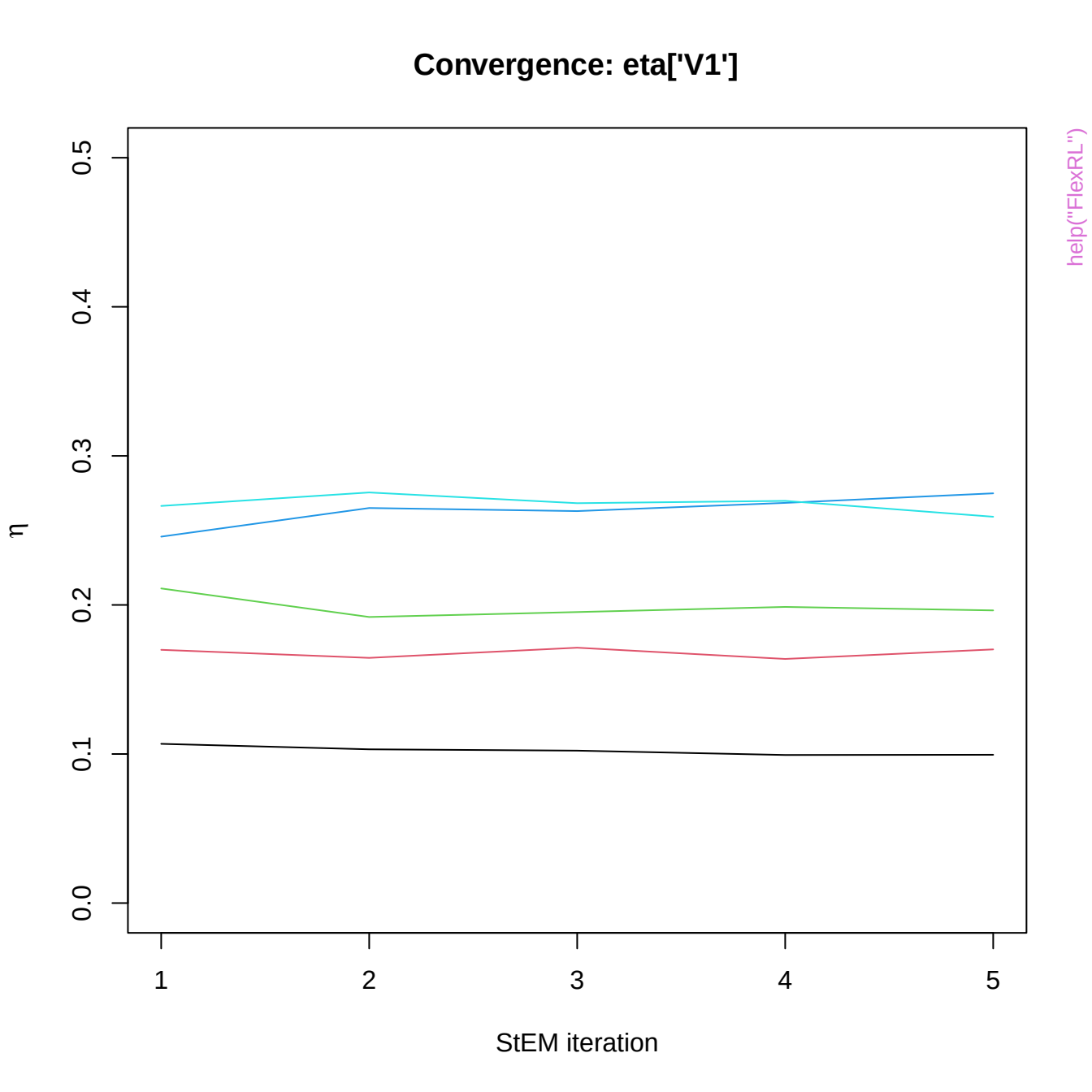

Distribution of V2
B: obs. 200
linkedB (0.75): obs. 81
Density
0.0
0.2
0.4
0.6
0.8
1.0
0
1
2
3
4
5
6
V2

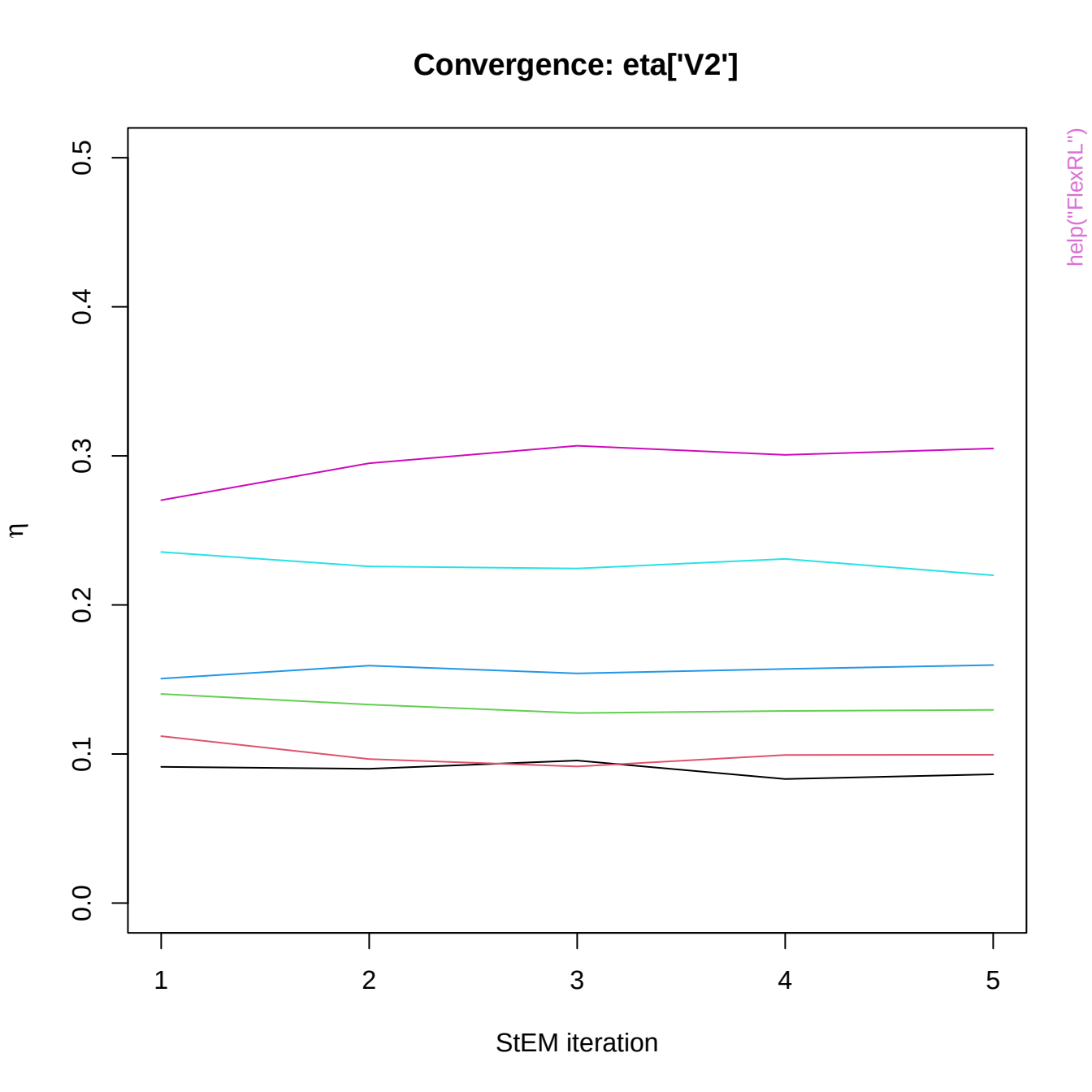

Distribution of V3
B: obs. 200
linkedB (0.75): obs. 81
Density
0.0
0.2
0.4
0.6
0.8
1.0
1
2
3
4
5
6
7
V3

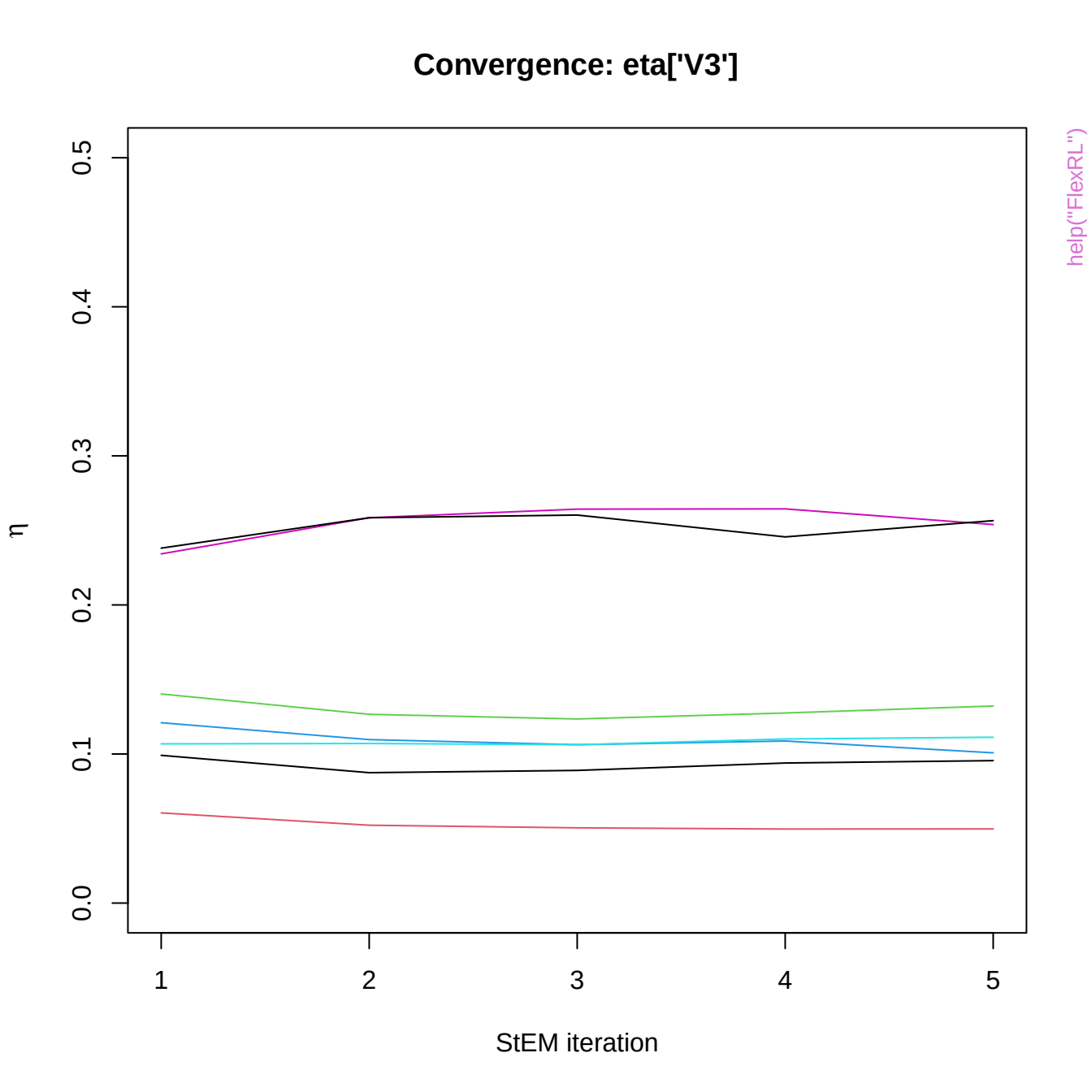

Distribution of V4
B: obs. 200
linkedB (0.75): obs. 81
Density
0.0
0.2
0.4
0.6
0.8
1.0
0
2
4
6
8
10
12
V4

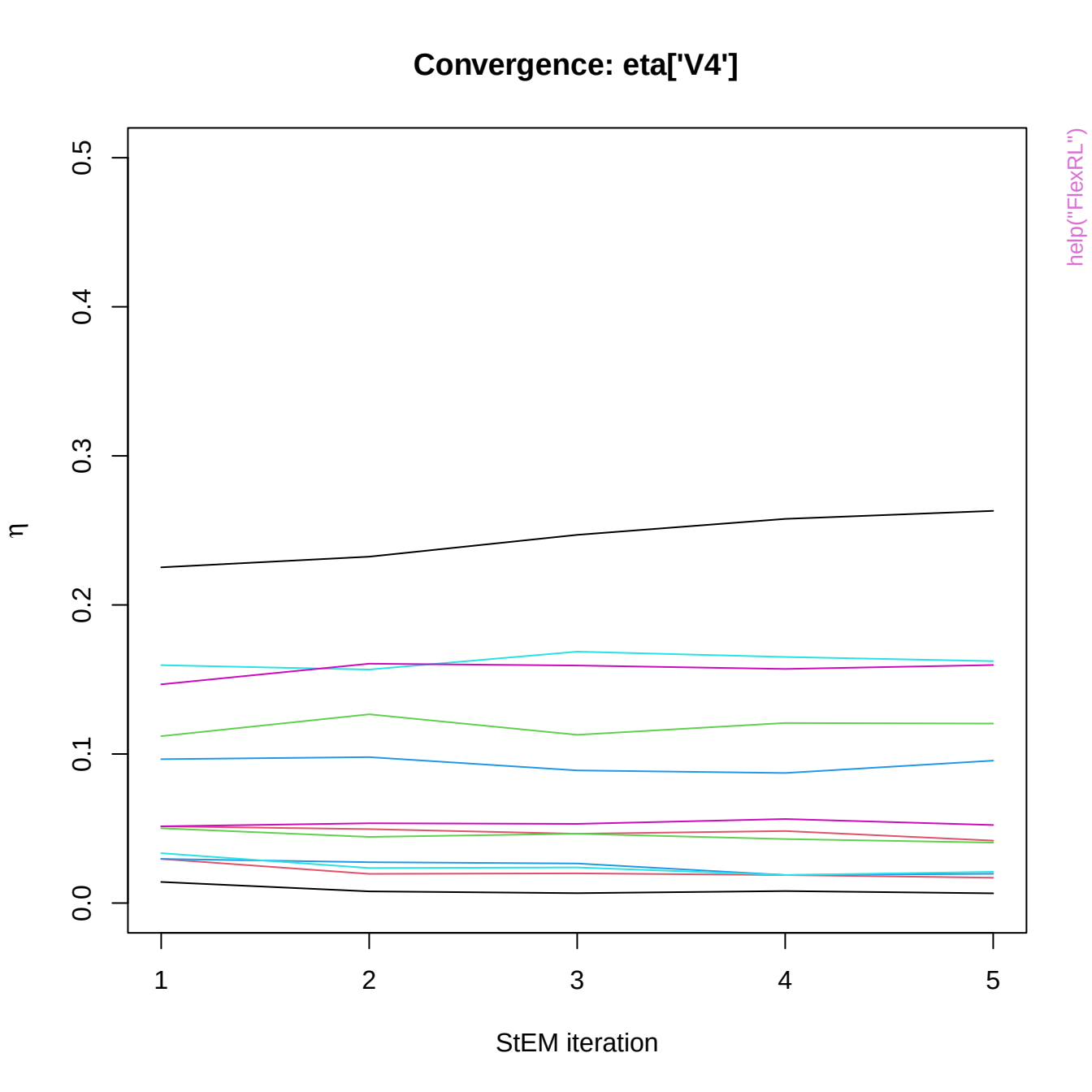
help("FlexRL")

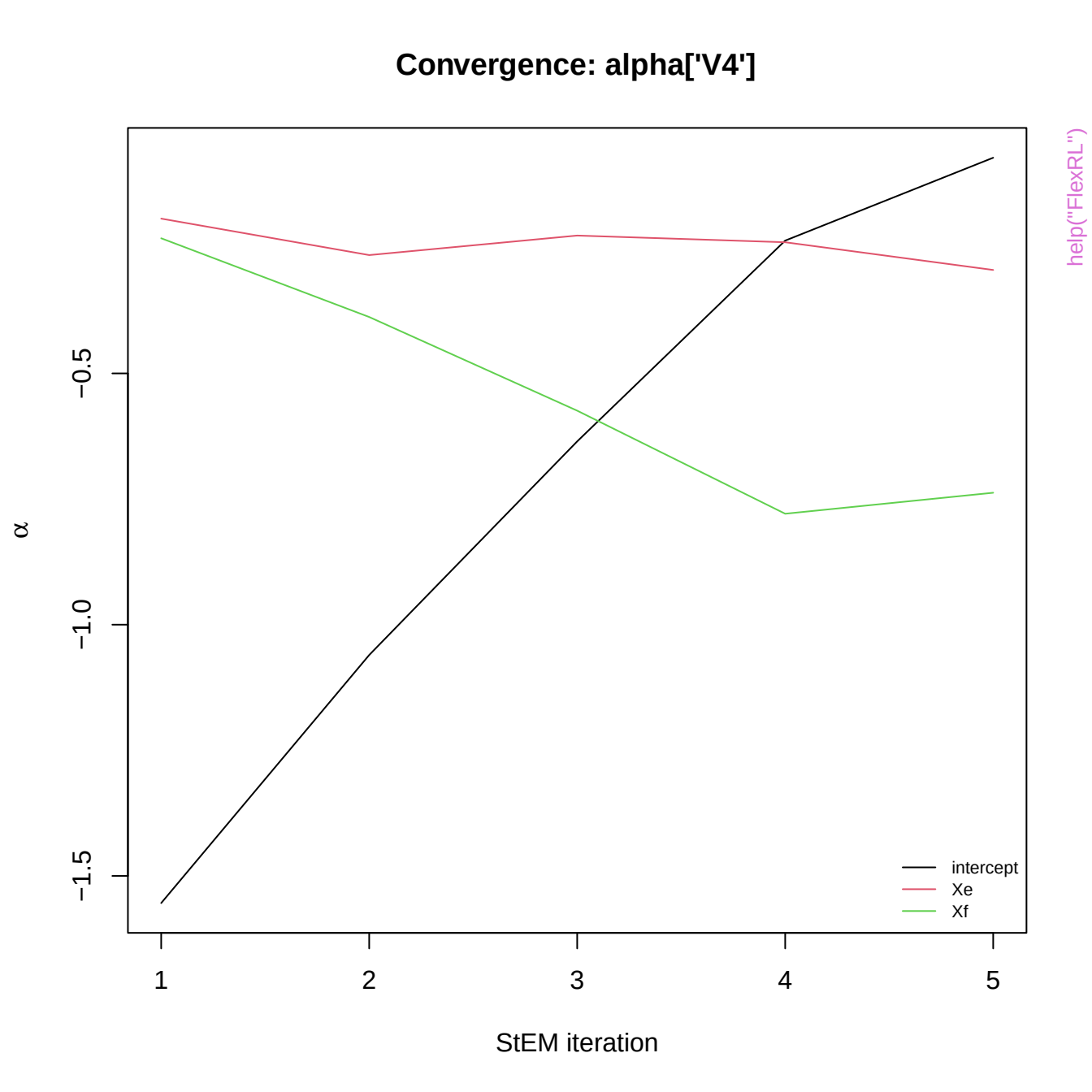

Convergence: gamma (proportion linked)
γ
0.61
0.62
0.63
0.64
0.65
0.66
0.67
0.68
1
2
3
4
5
StEM iteration
help("FlexRL")

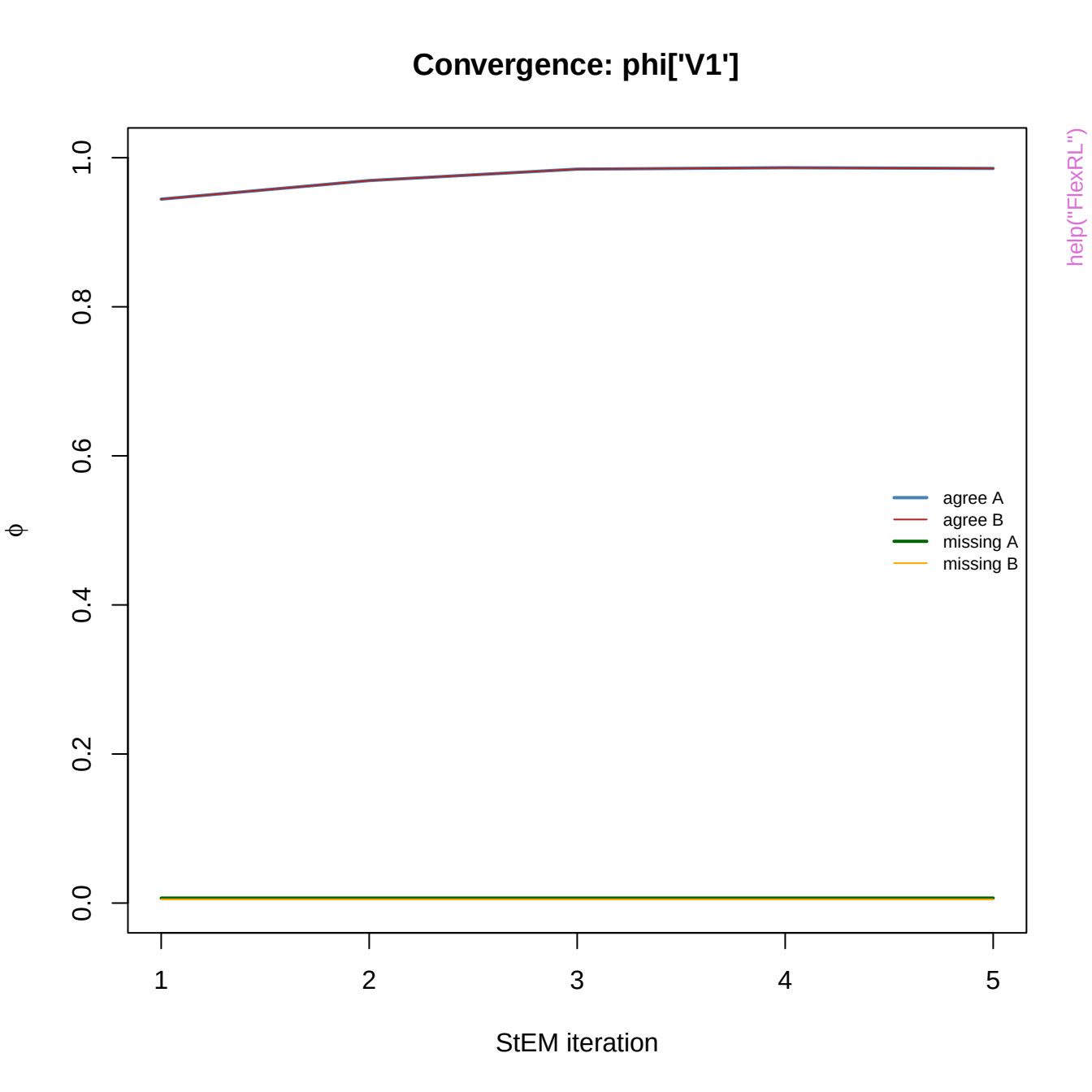

Convergence: eta['V1']
η
0.0
0.1
0.2
0.3
0.4
0.5
1
2
3
4
5
StEM iteration
help("FlexRL")

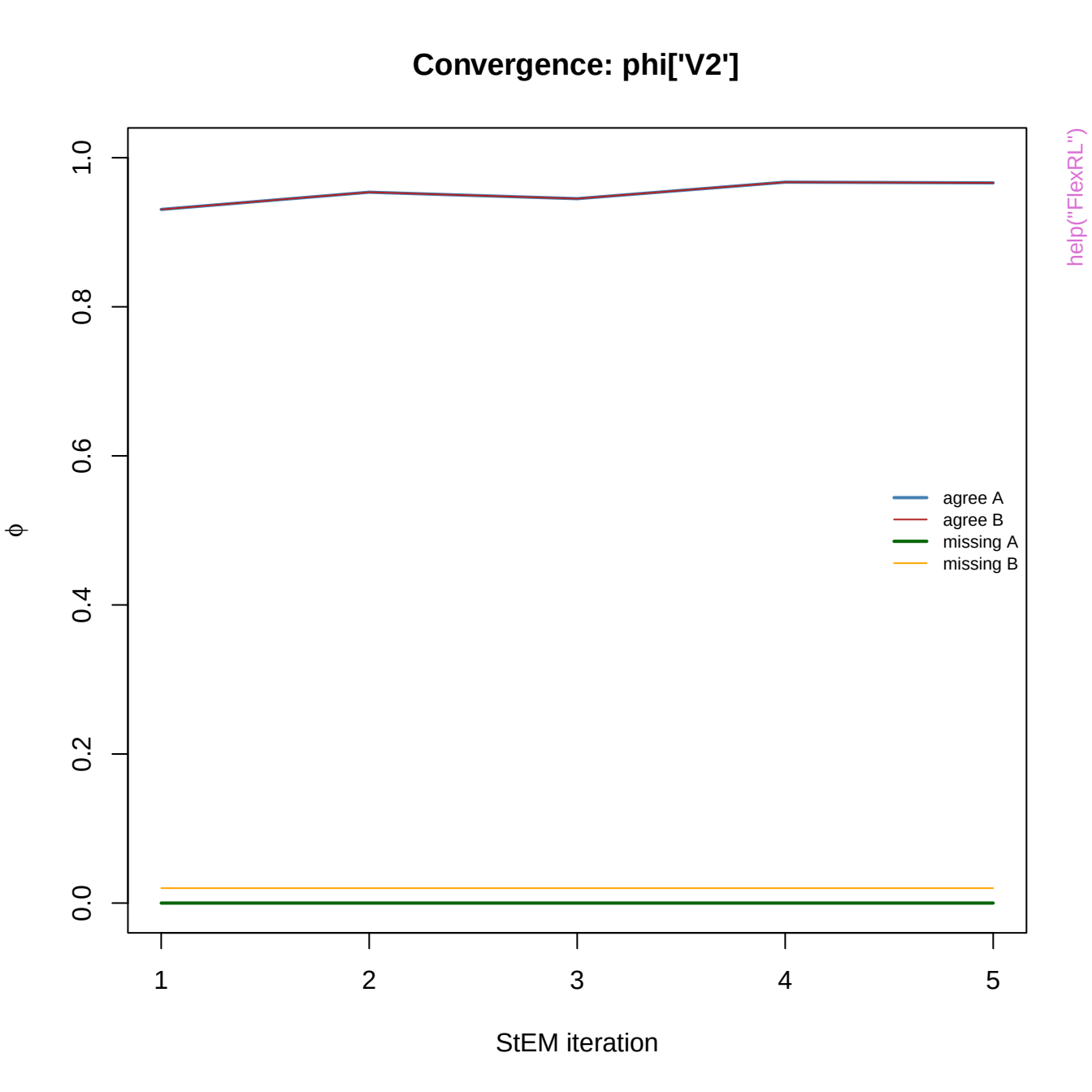

Convergence: eta['V2']
η
0.0
0.1
0.2
0.3
0.4
0.5
1
2
3
4
5
StEM iteration
help("FlexRL")

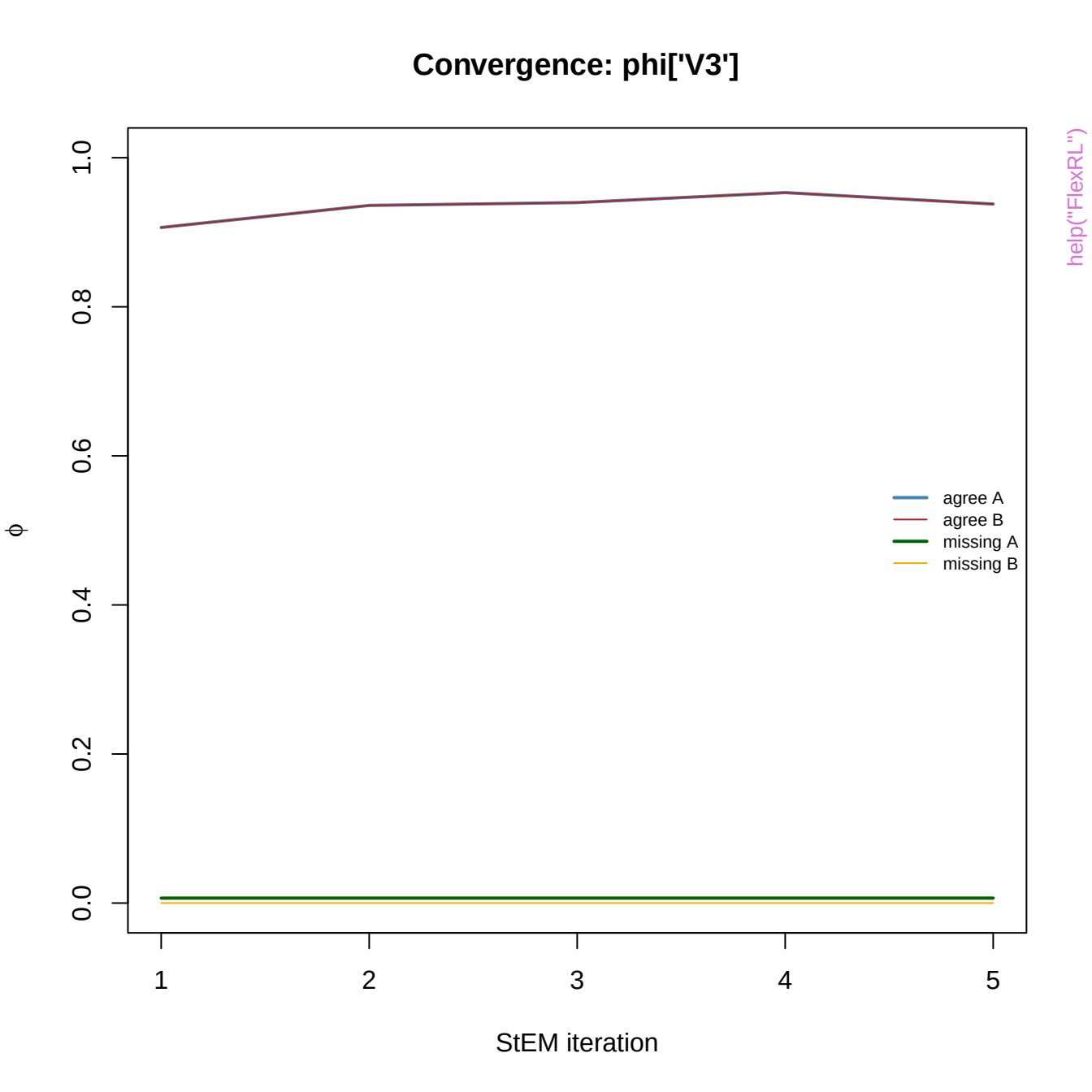

Convergence: eta['V3']
η
0.0
0.1
0.2
0.3
0.4
0.5
1
2
3
4
5
StEM iteration
help("FlexRL")

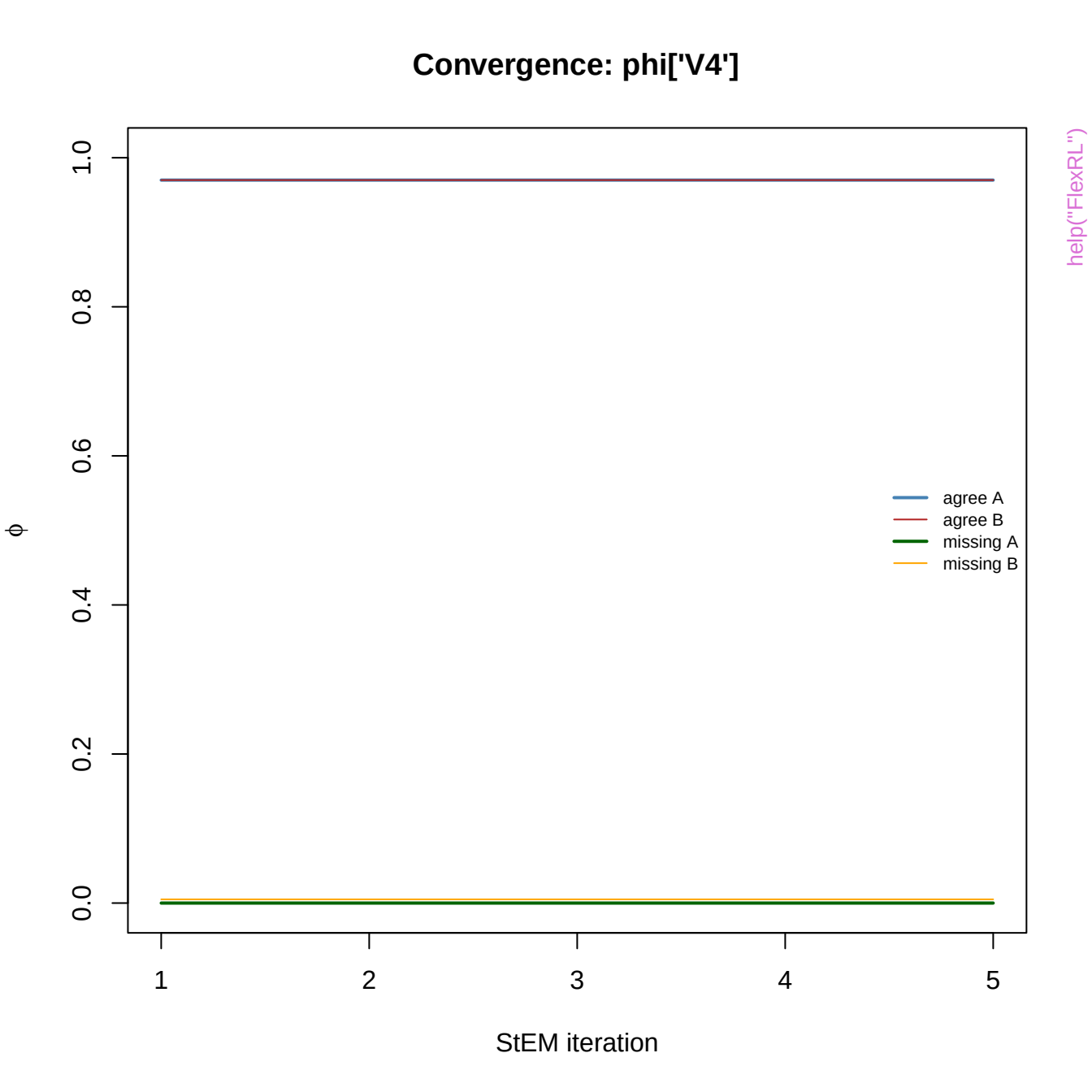

Convergence: eta['V4']
η
0.0
0.1
0.2
0.3
0.4
0.5
1
2
3
4
5
StEM iteration
help("FlexRL")

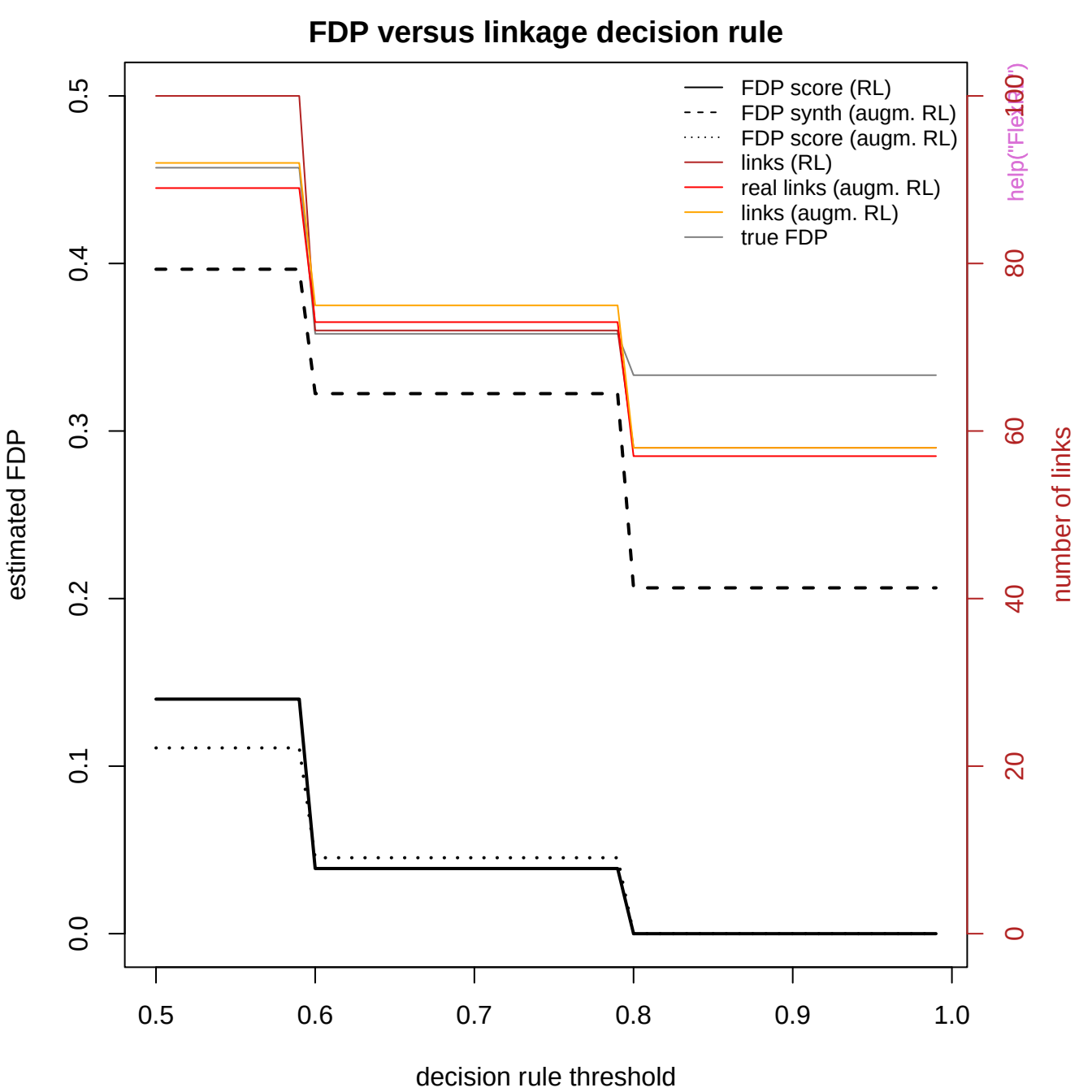

Convergence: alpha['V4']
α
−0.5
−1.0
−1.5
1
2
3
4
5
StEM iteration
intercept
Xe
Xf
help("FlexRL")

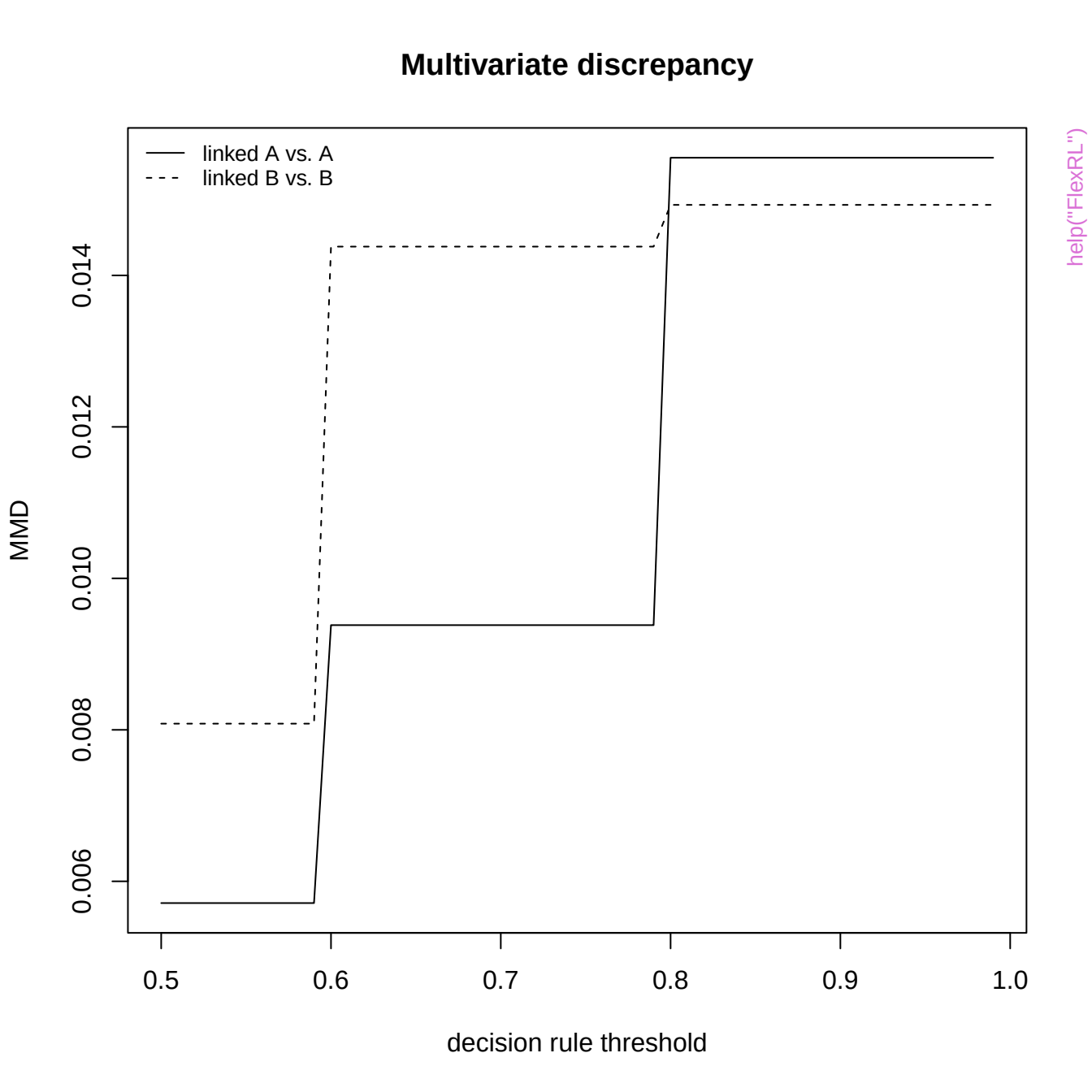

Convergence: phi['V1']
1.0
0.8
0.6
0.4
0.2
0.0
ϕ
agree A
agree B
missing A
missing B
1
2
3
4
5
StEM iteration
help("FlexRL")

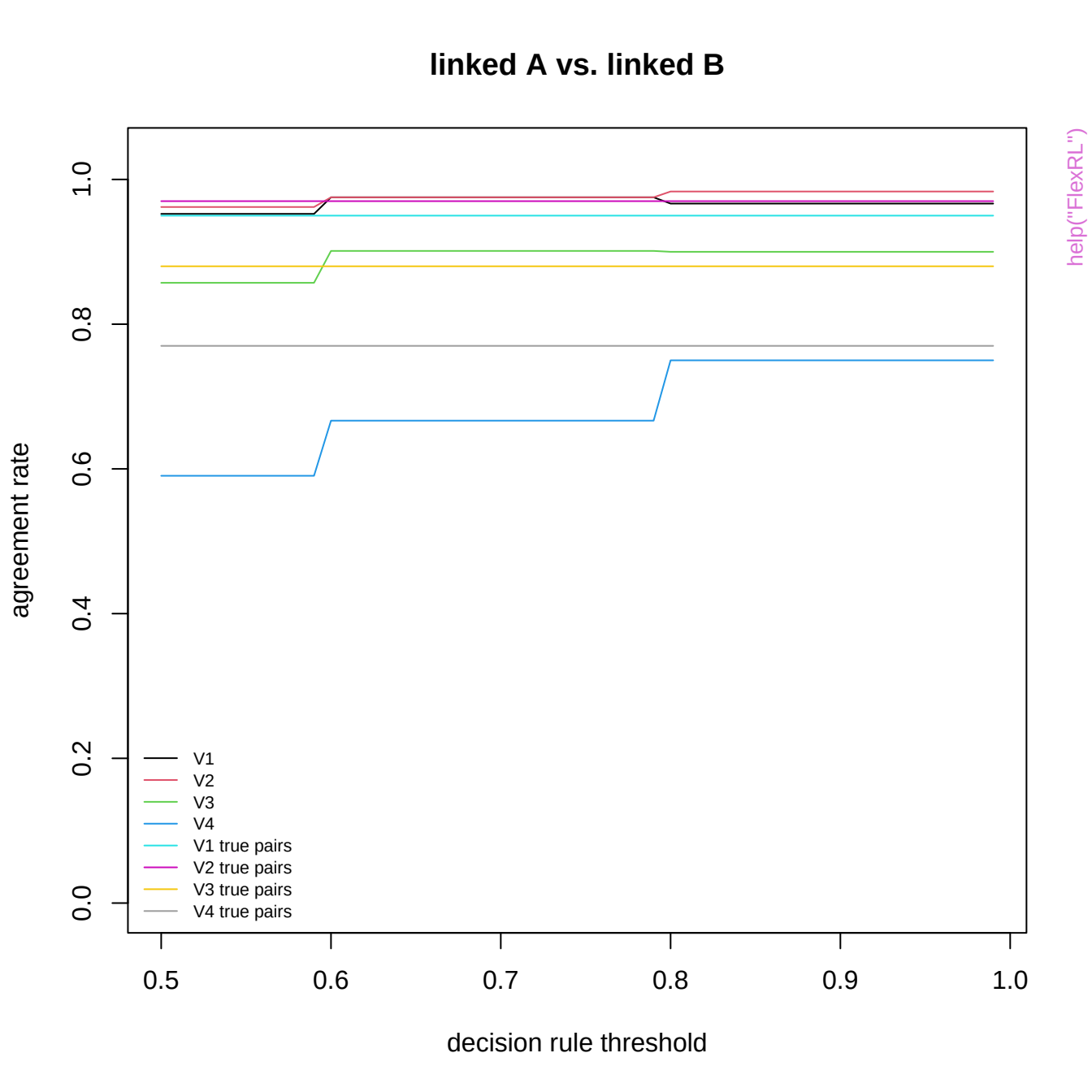

Convergence: phi['V2']
1.0
0.8
0.6
0.4
0.2
0.0
ϕ
agree A
agree B
missing A
missing B
1
2
3
4
5
StEM iteration
help("FlexRL")

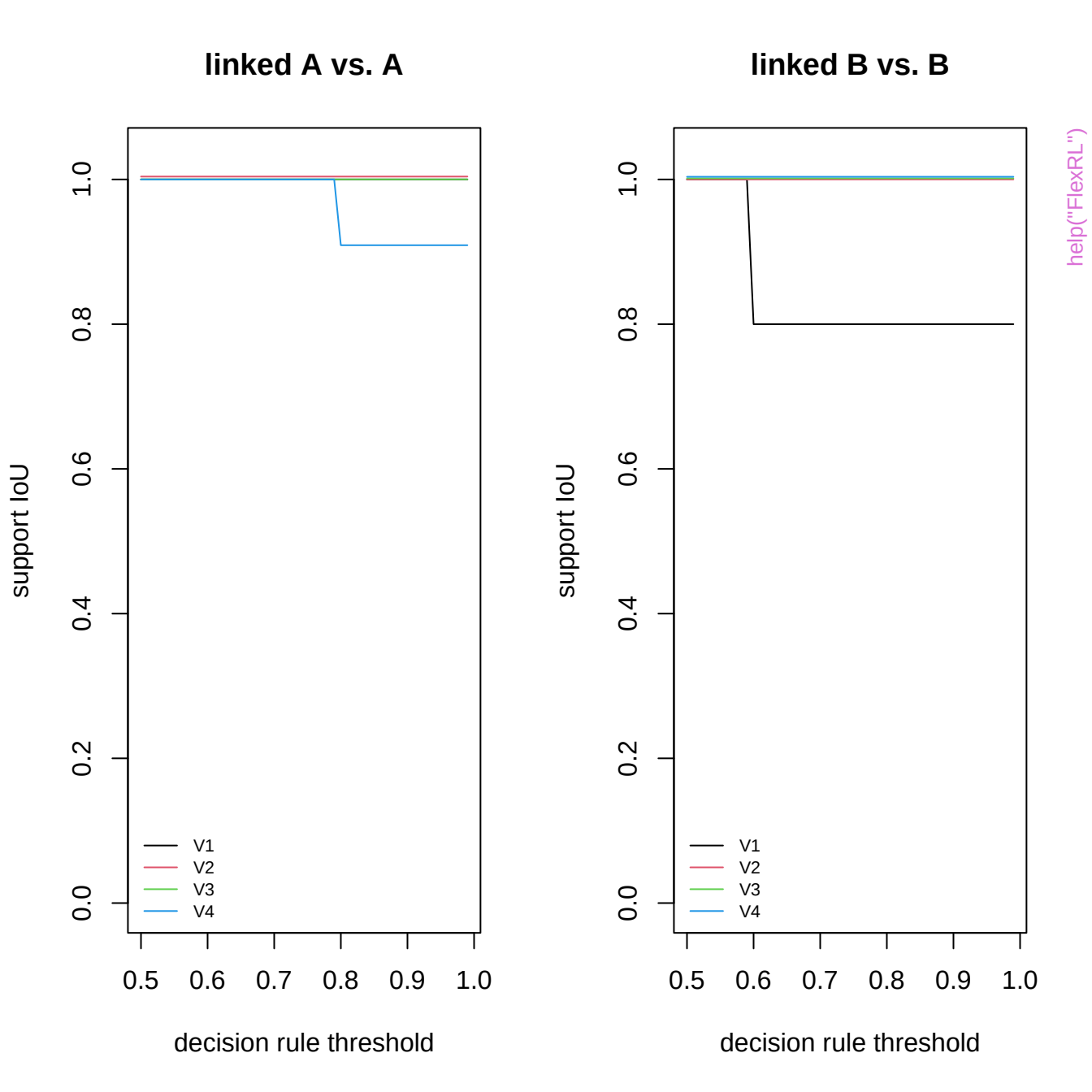

Convergence: phi['V3']
1.0
0.8
0.6
0.4
0.2
0.0
ϕ
agree A
agree B
missing A
missing B
1
2
3
4
5
StEM iteration
help("FlexRL")

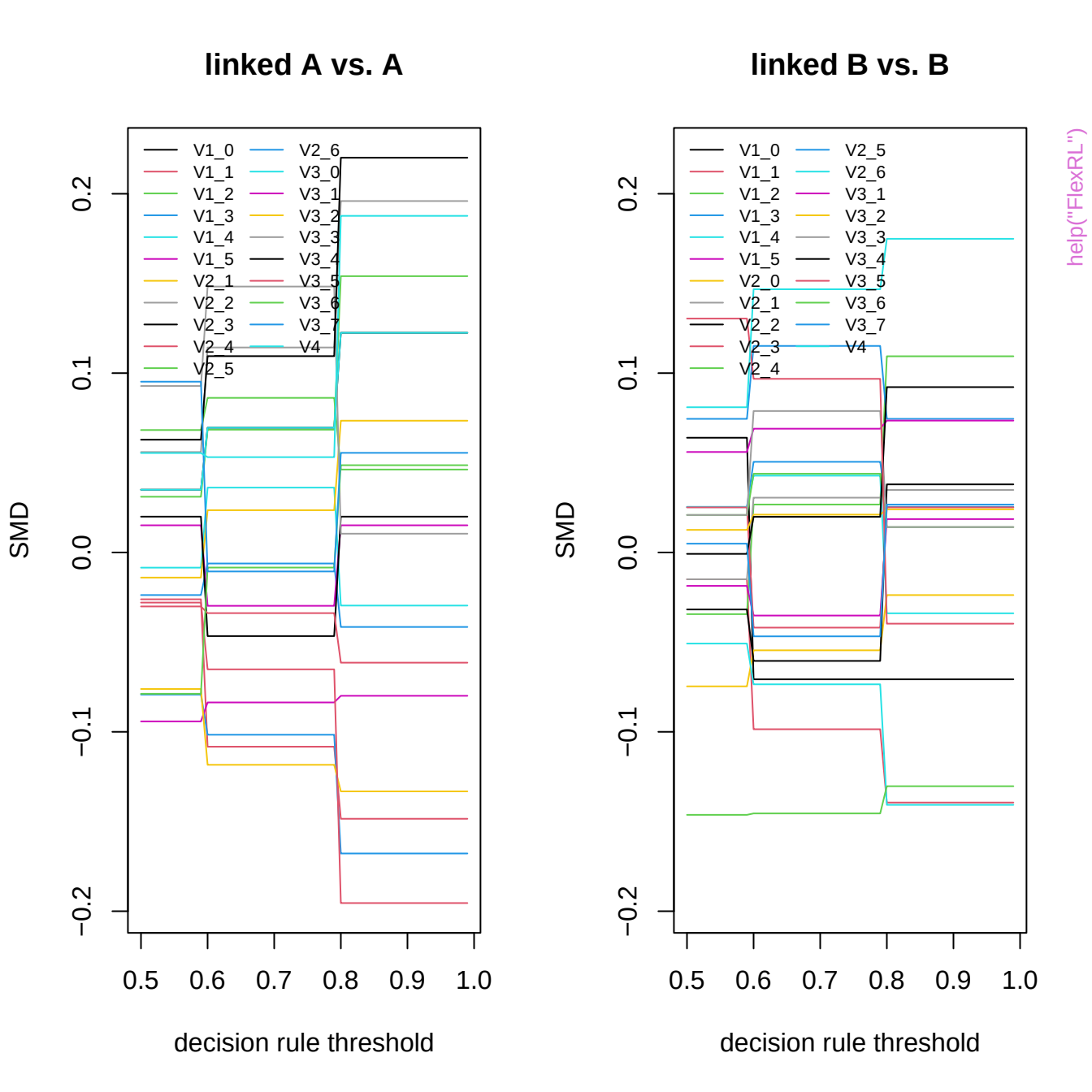

Convergence: phi['V4']
1.0
0.8
0.6
0.4
0.2
0.0
ϕ
agree A
agree B
missing A
missing B
1
2
3
4
5
StEM iteration
help("FlexRL")

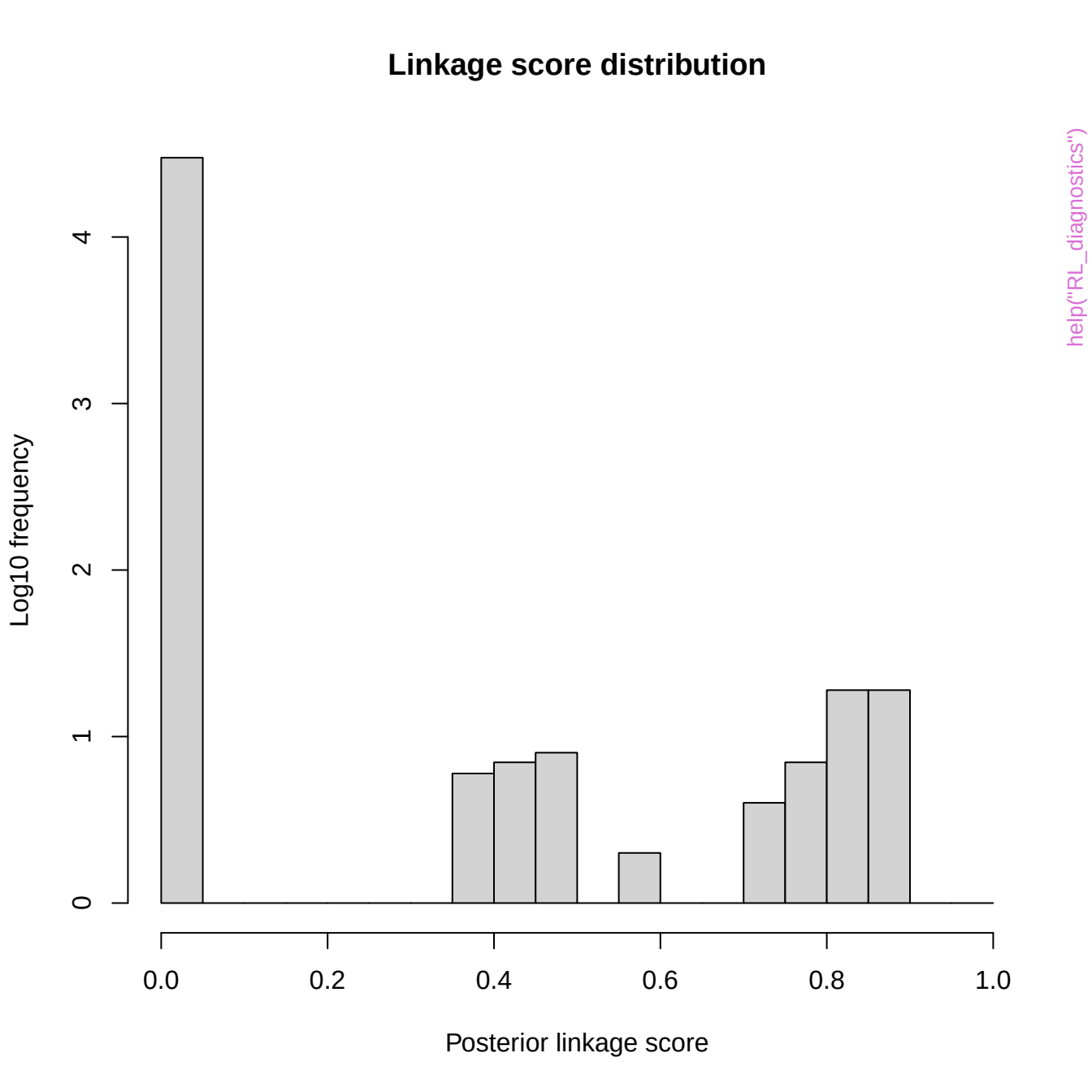

FDP versus linkage decision rule
FDP score (RL)
FDP synth (augm. RL)
FDP score (augm. RL)
links (RL)
real links (augm. RL)
links (augm. RL)
true FDP
estimated FDP
number of links
decision rule threshold
0.0
0.1
0.2
0.3
0.4
0.5
0
20
40
60
80
100
0.5
0.6
0.7
0.8
0.9
1.0

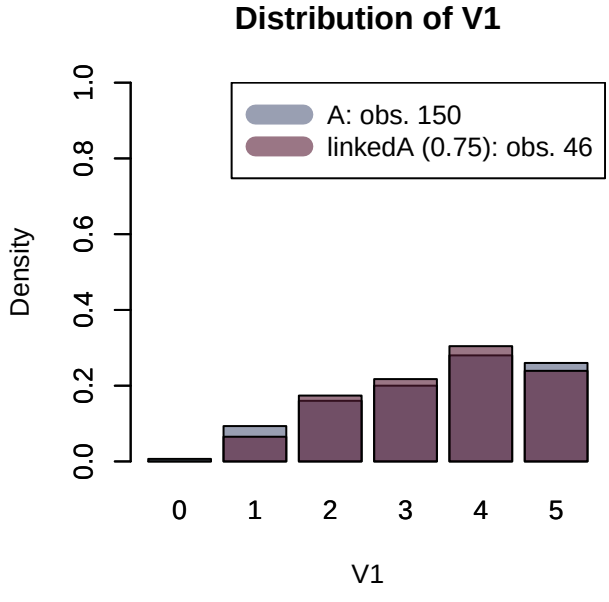

Multivariate discrepancy
linked A vs. A
linked B vs. B
MMD
0.006
0.008
0.010
0.012
0.014
0.5
0.6
0.7
0.8
0.9
1.0
decision rule threshold
help("FlexRL")

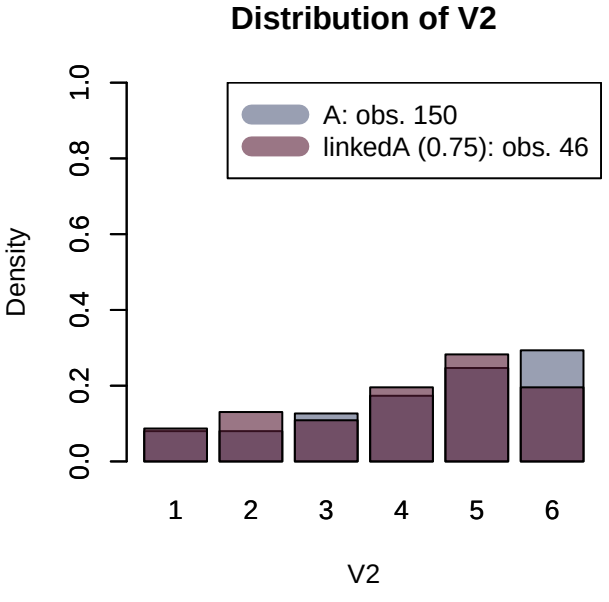

linked A vs. linked B
agreement rate
0.0
0.2
0.4
0.6
0.8
1.0
0.5
0.6
0.7
0.8
0.9
1.0
decision rule threshold
V1
V2
V3
V4
V1 true pairs
V2 true pairs
V3 true pairs
V4 true pairs
help("FlexRL")

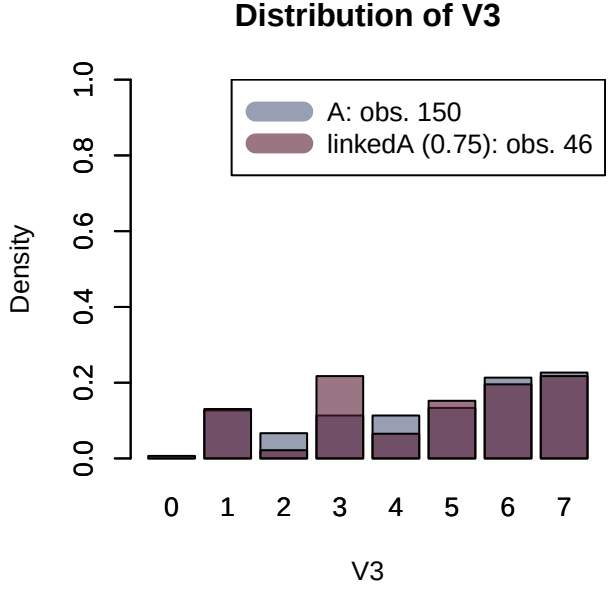

linked A vs. A
linked B vs. B
support IoU
decision rule threshold
0.0
0.2
0.4
0.6
0.8
1.0
0.5
0.6
0.7
0.8
0.9
1.0
V1
V2
V3
V4
help("FlexRL")

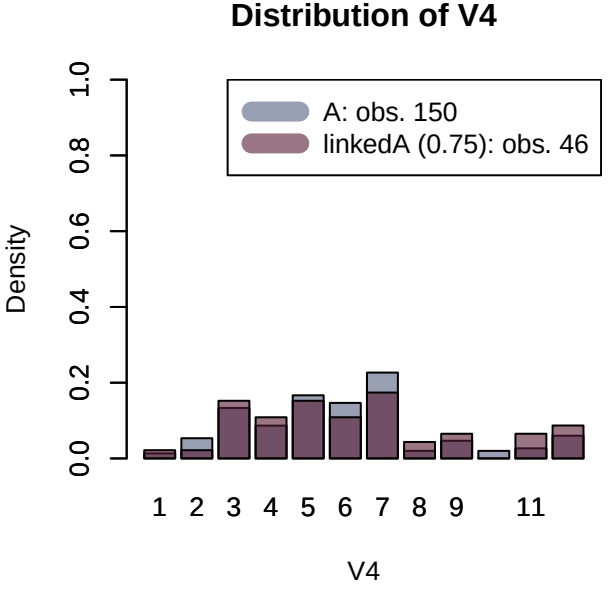

linked A vs. A
linked B vs. B
SMD
decision rule threshold
V1_0
V1_1
V1_2
V1_3
V1_4
V1_5
V2_0
V2_1
V2_2
V2_3
V2_4
V2_5
V2_6
V3_0
V3_1
V3_2
V3_3
V3_4
V3_5
V3_6
V3_7
V4
help("FlexRL")


Linkage score distribution
help("RL_diagnostics")
Log10 frequency
0
1
2
3
4
0.0
0.2
0.4
0.6
0.8
1.0
Posterior linkage score

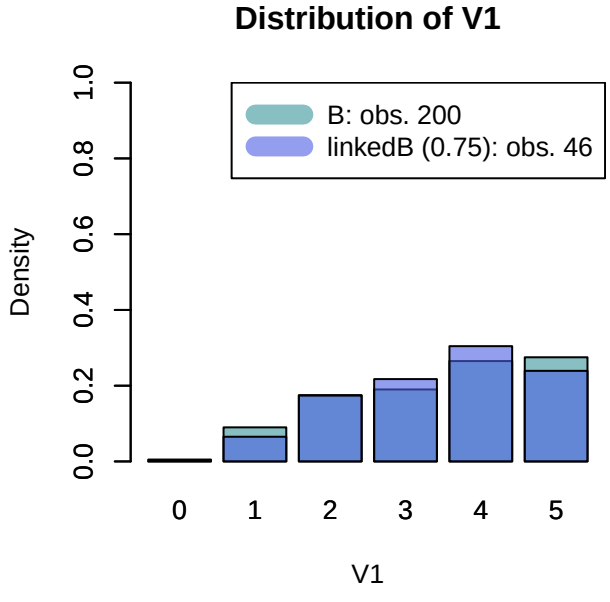


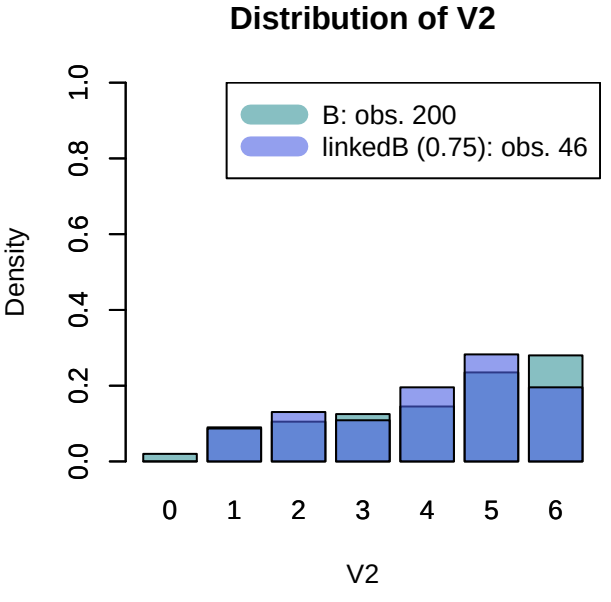


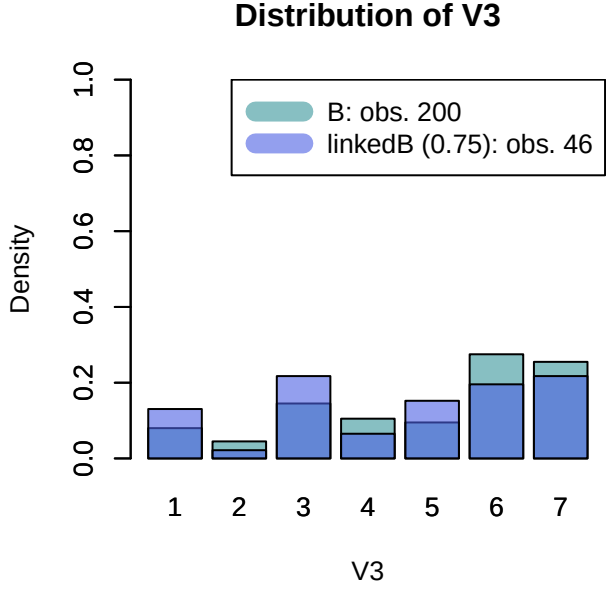


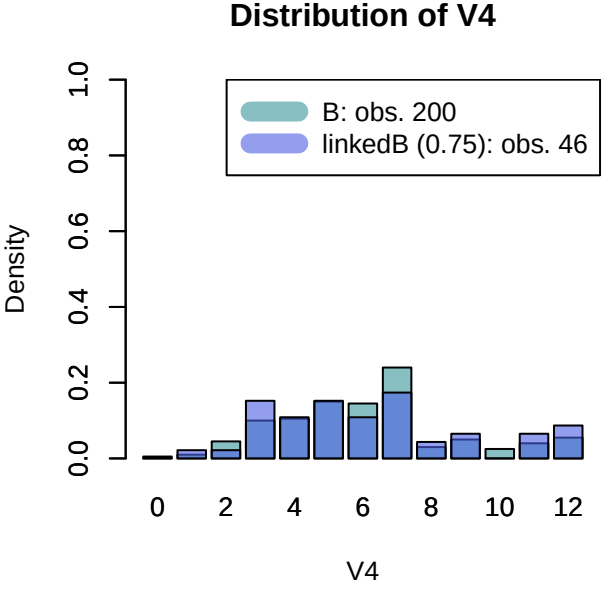


help("RL_diagnostics")

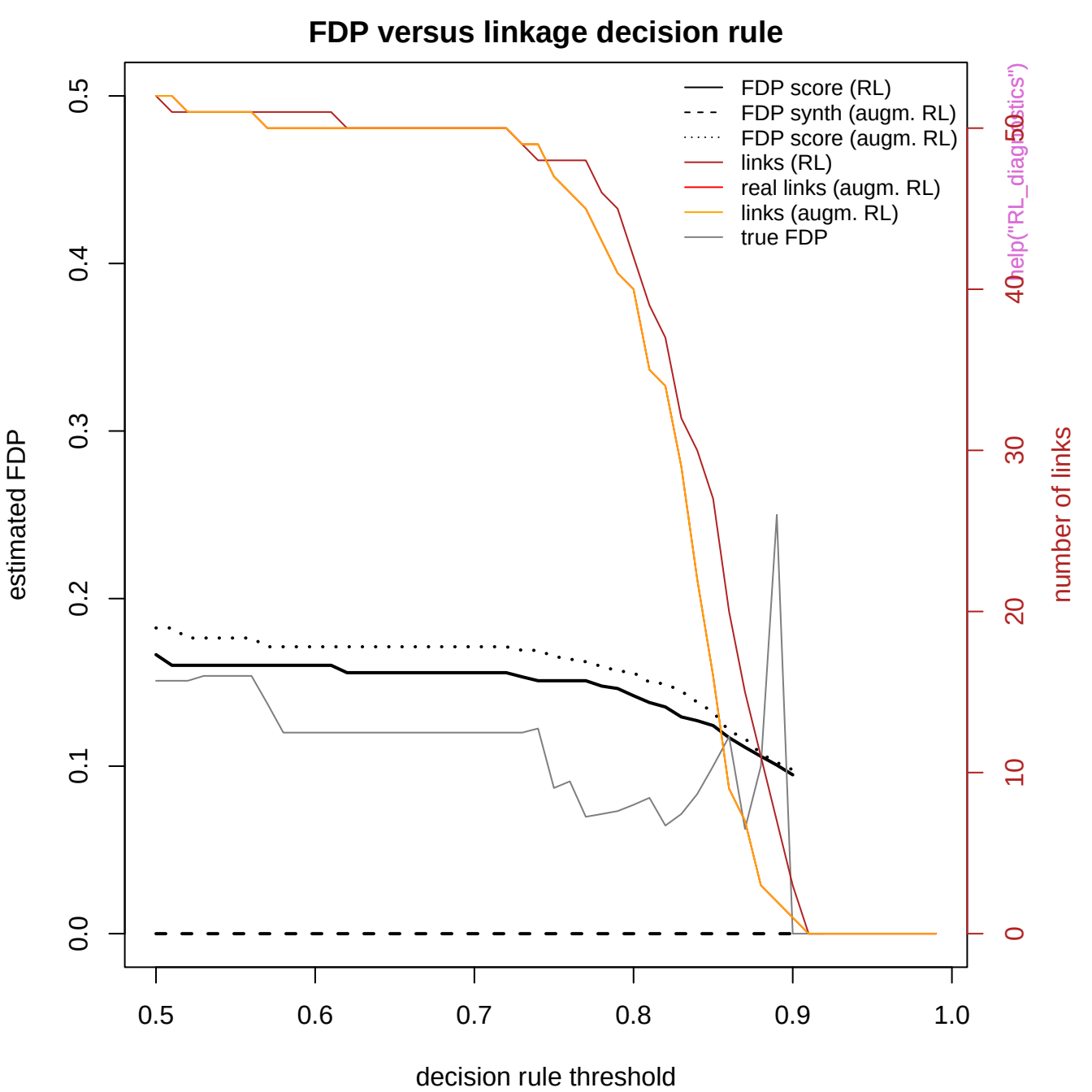


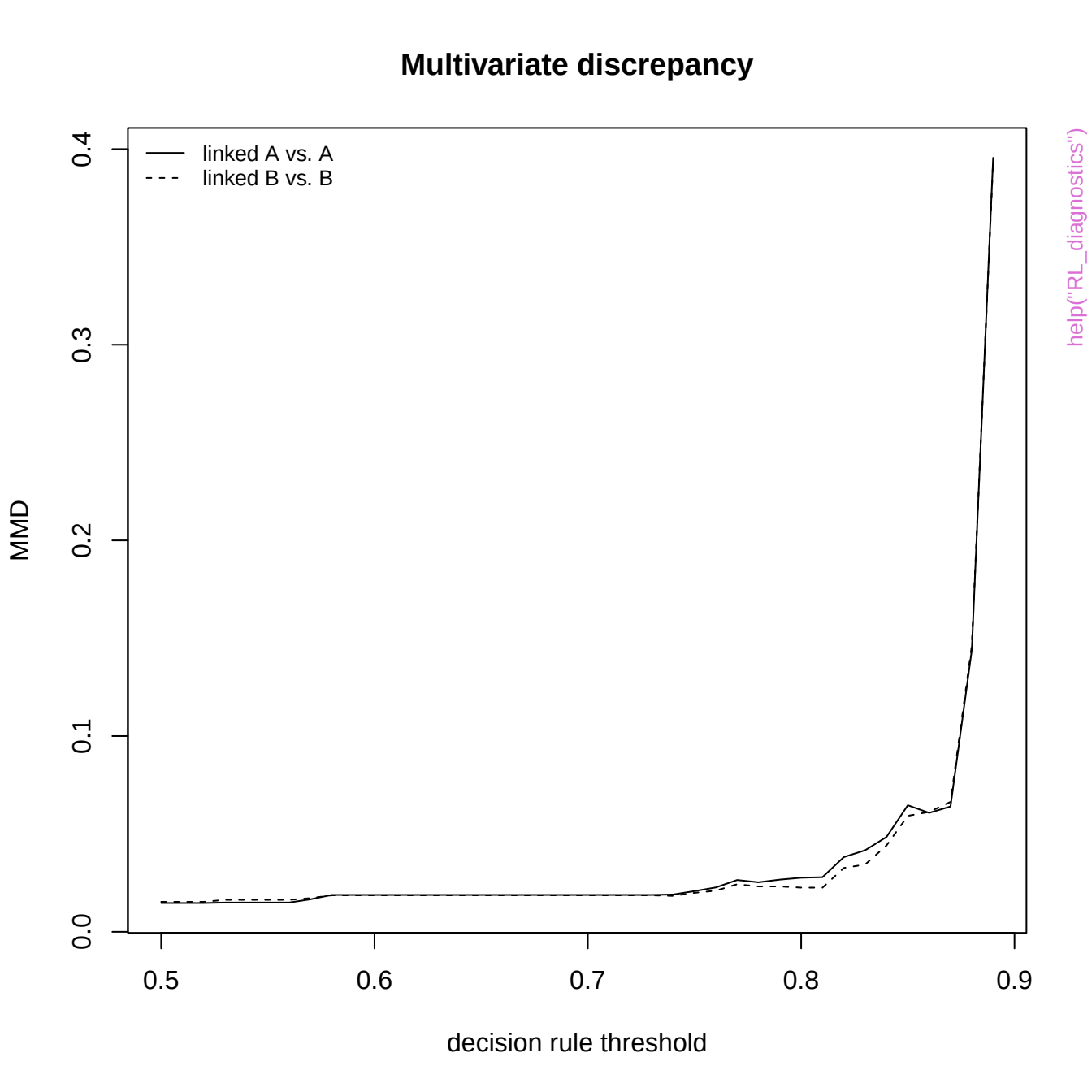


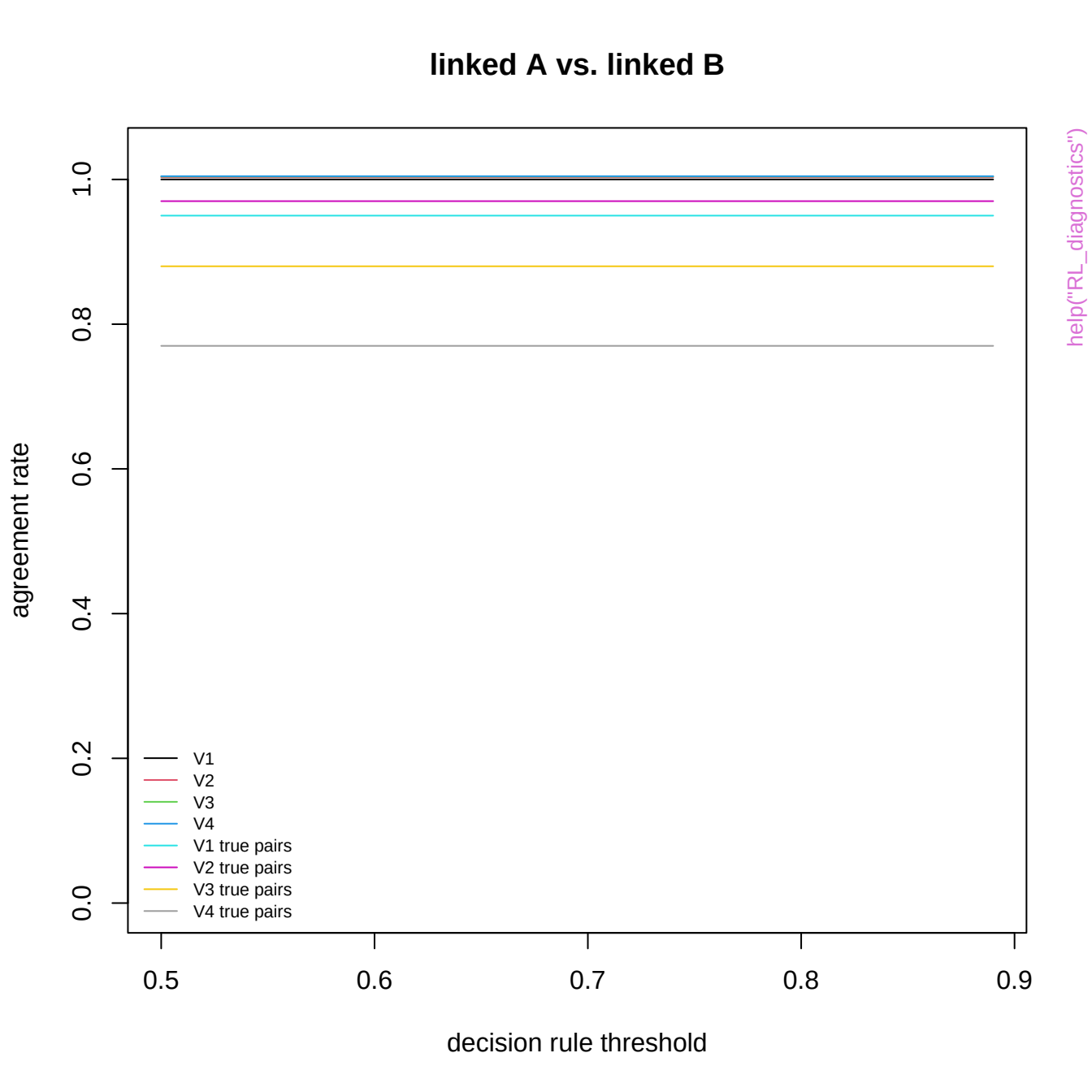


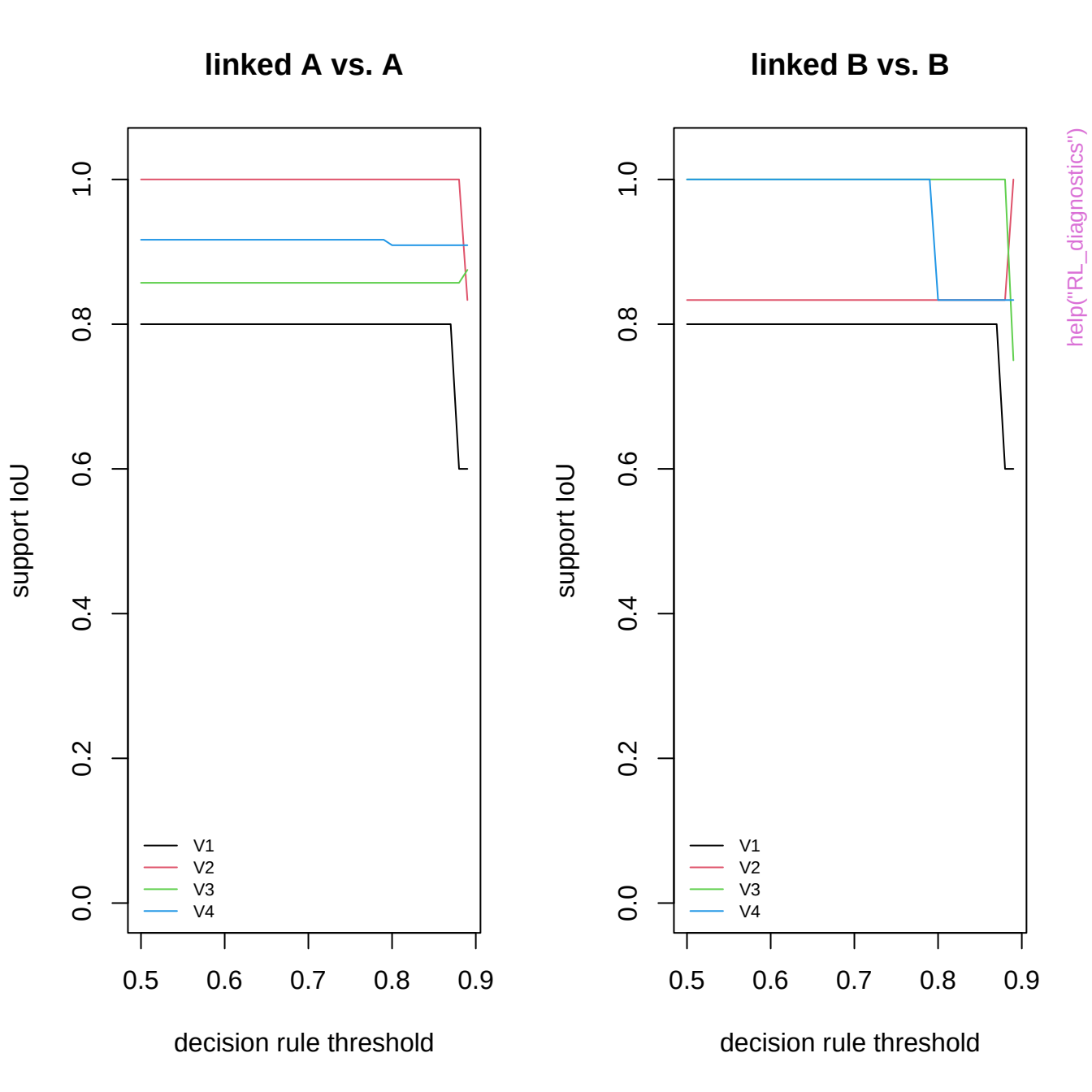


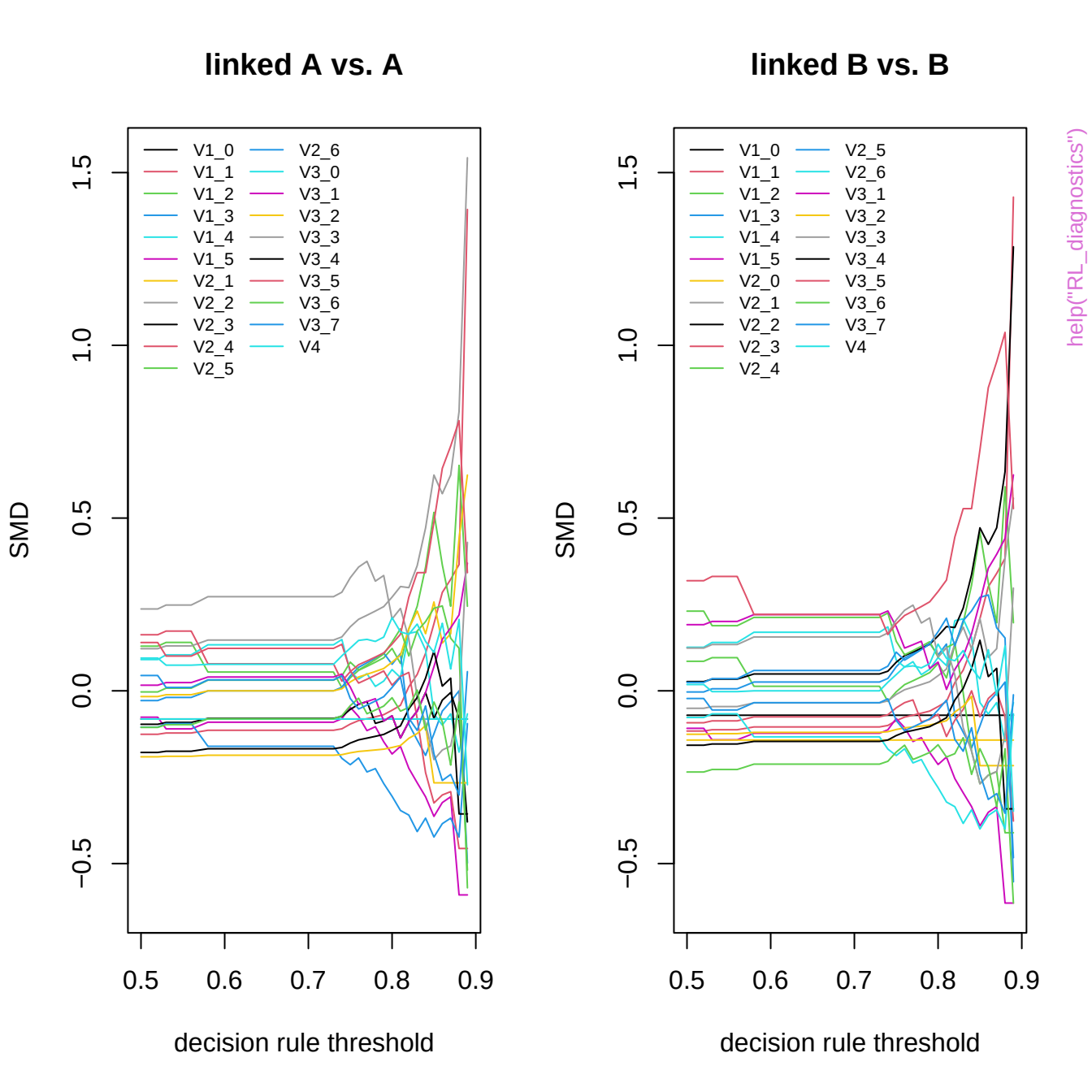
help("RL_diagnostics")

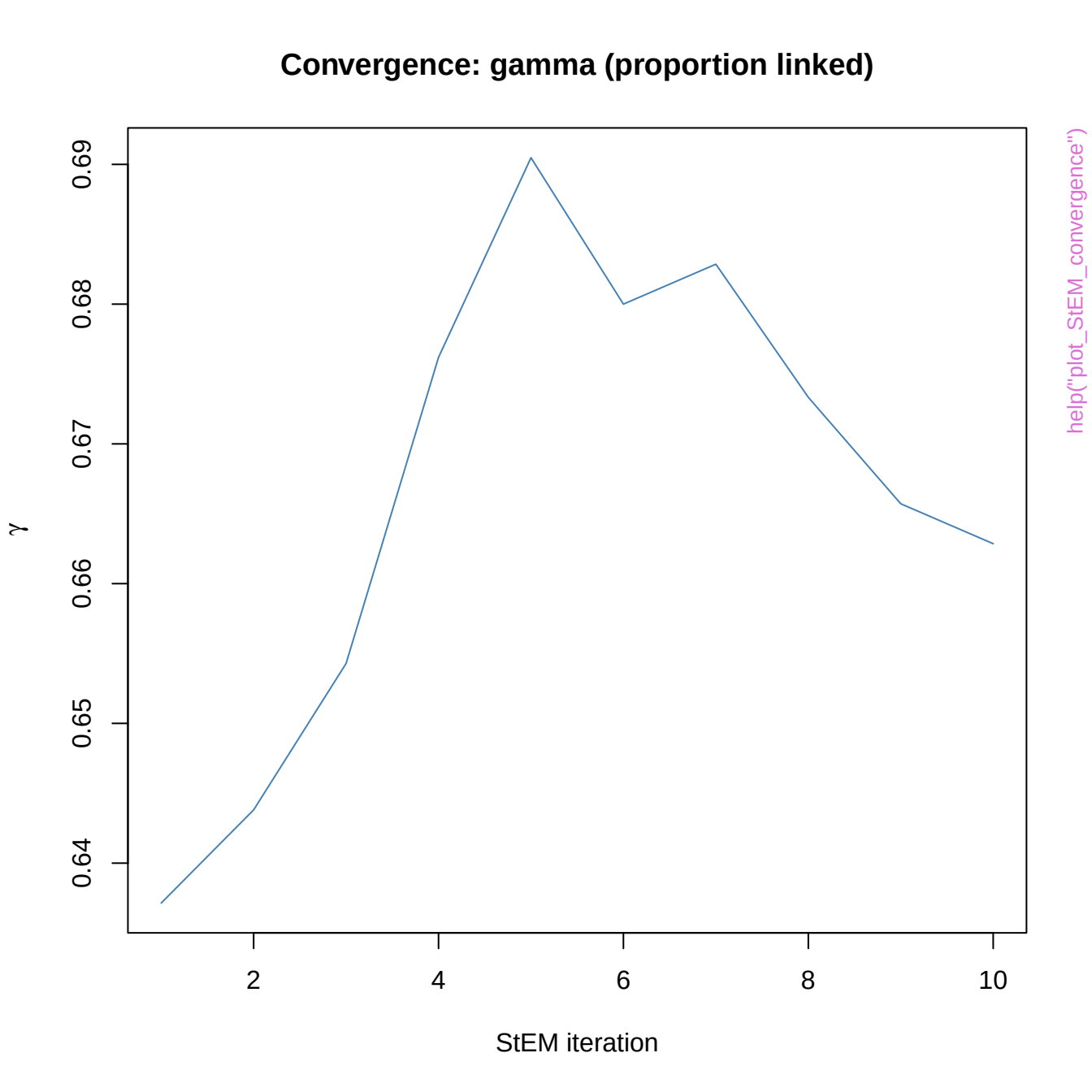

FDP versus linkage decision rule
FDP score (RL)
FDP synth (augm. RL)
FDP score (augm. RL)
links (RL)
real links (augm. RL)
links (augm. RL)
true FDP
estimated FDP
number of links
decision rule threshold
0.0
0.1
0.2
0.3
0.4
0.5
0
10
20
30
40
50
0.5
0.6
0.7
0.8
0.9
1.0

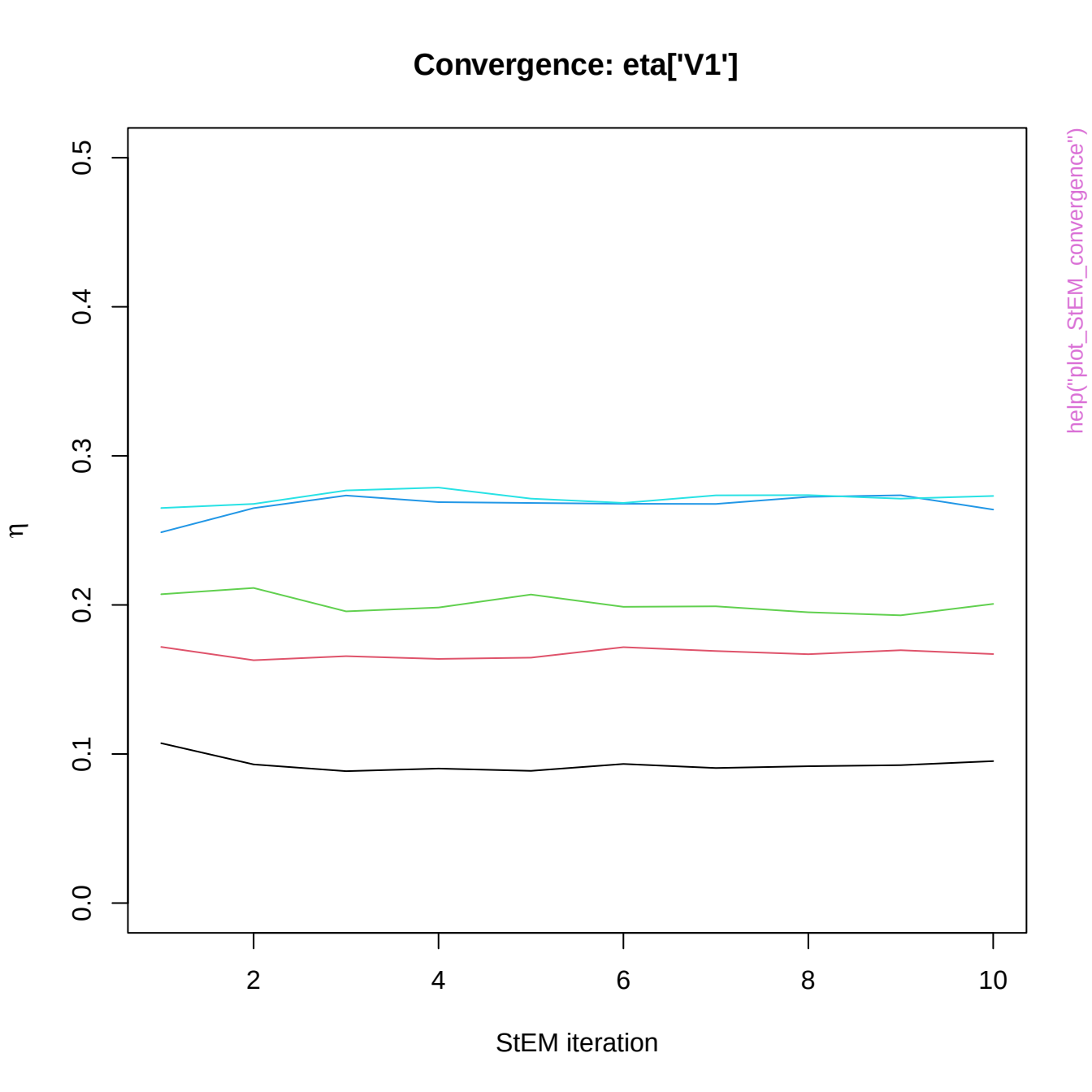

Multivariate discrepancy
linked A vs. A
linked B vs. B
MMD
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
decision rule threshold
help("RL_diagnostics")

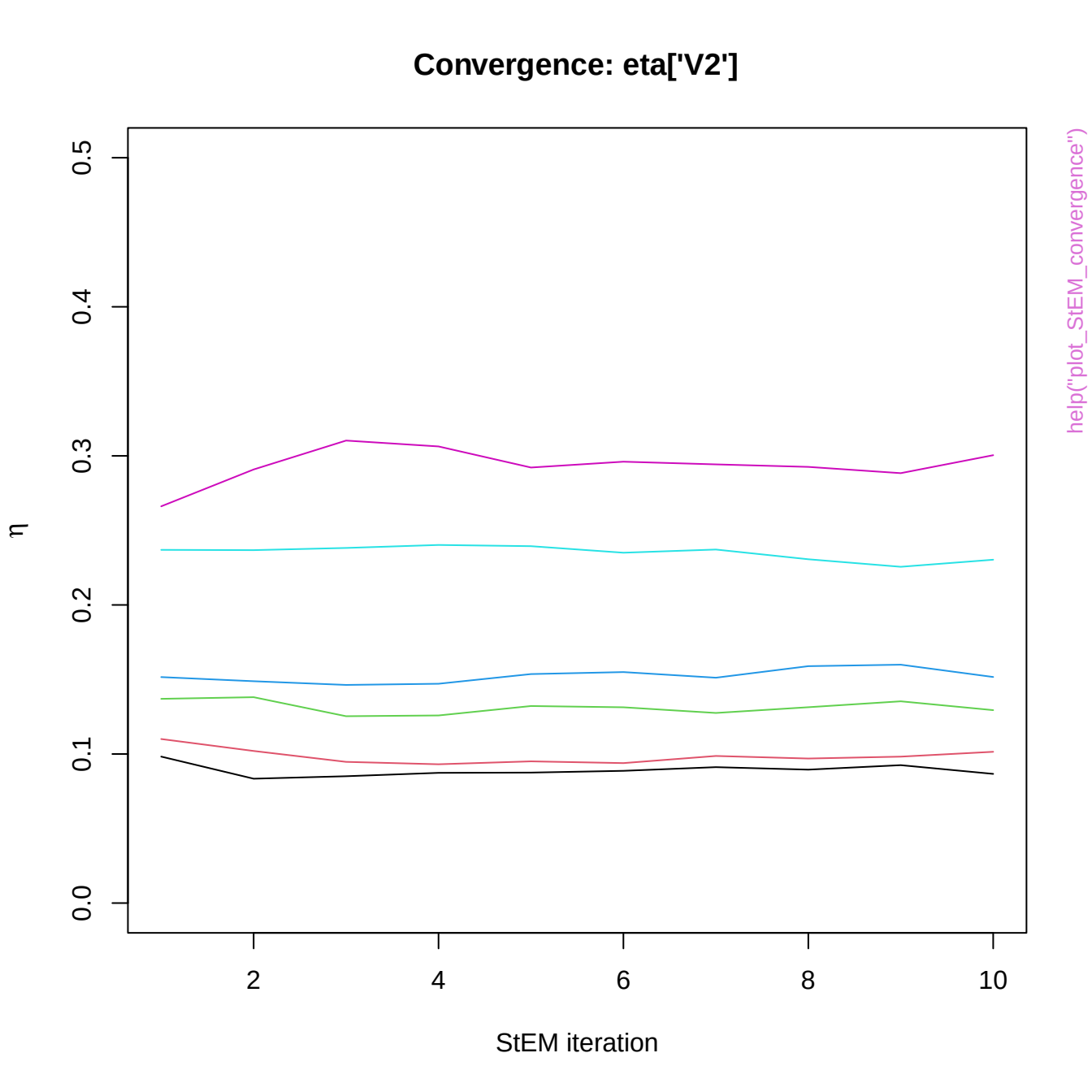

linked A vs. linked B
agreement rate
1.0
0.8
0.6
0.4
0.2
0.0
0.5
0.6
0.7
0.8
0.9
decision rule threshold
V1
V2
V3
V4
V1 true pairs
V2 true pairs
V3 true pairs
V4 true pairs
help("RL_diagnostics")

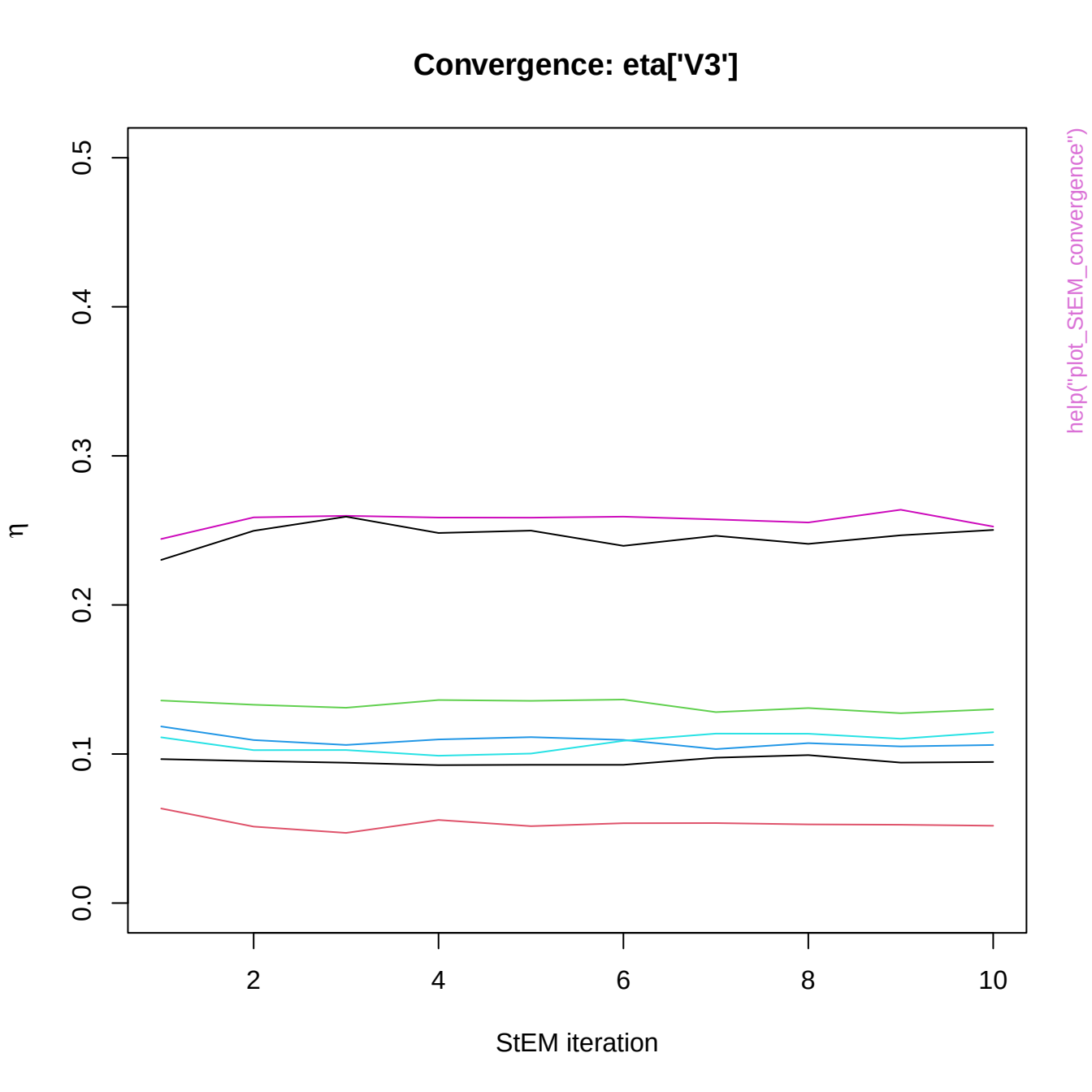

linked A vs. A
linked B vs. B
support IoU
1.0
0.8
0.6
0.4
0.2
0.0
0.5
0.6
0.7
0.8
0.9
decision rule threshold
V1
V2
V3
V4
help("RL_diagnostics")

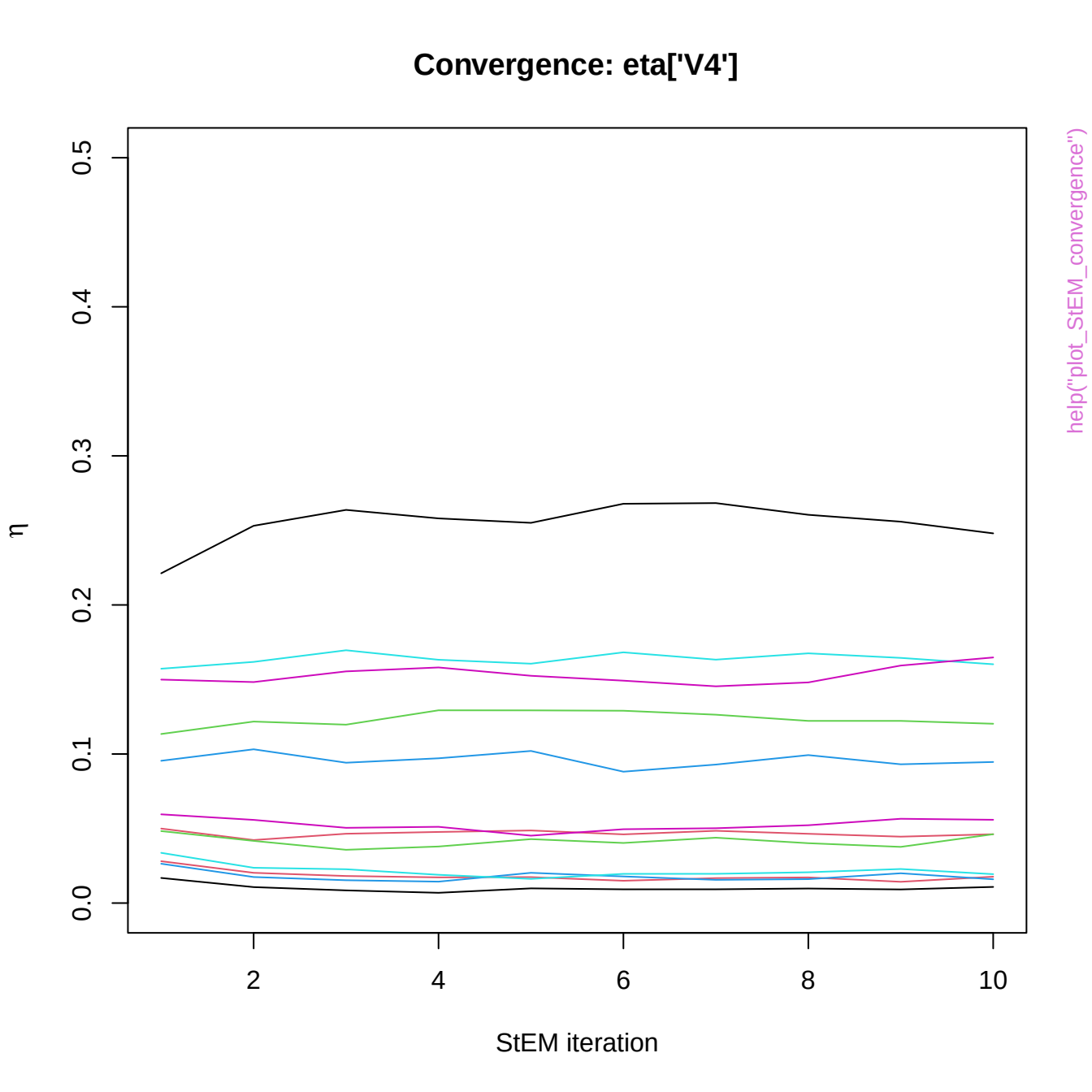

linked A vs. A
linked B vs. B
SMD
decision rule threshold
V1_0
V1_1
V1_2
V1_3
V1_4
V1_5
V2_0
V2_1
V2_2
V2_3
V2_4
V2_5
V2_6
V3_0
V3_1
V3_2
V3_3
V3_4
V3_5
V3_6
V3_7
V4
help("RL_diagnostics")

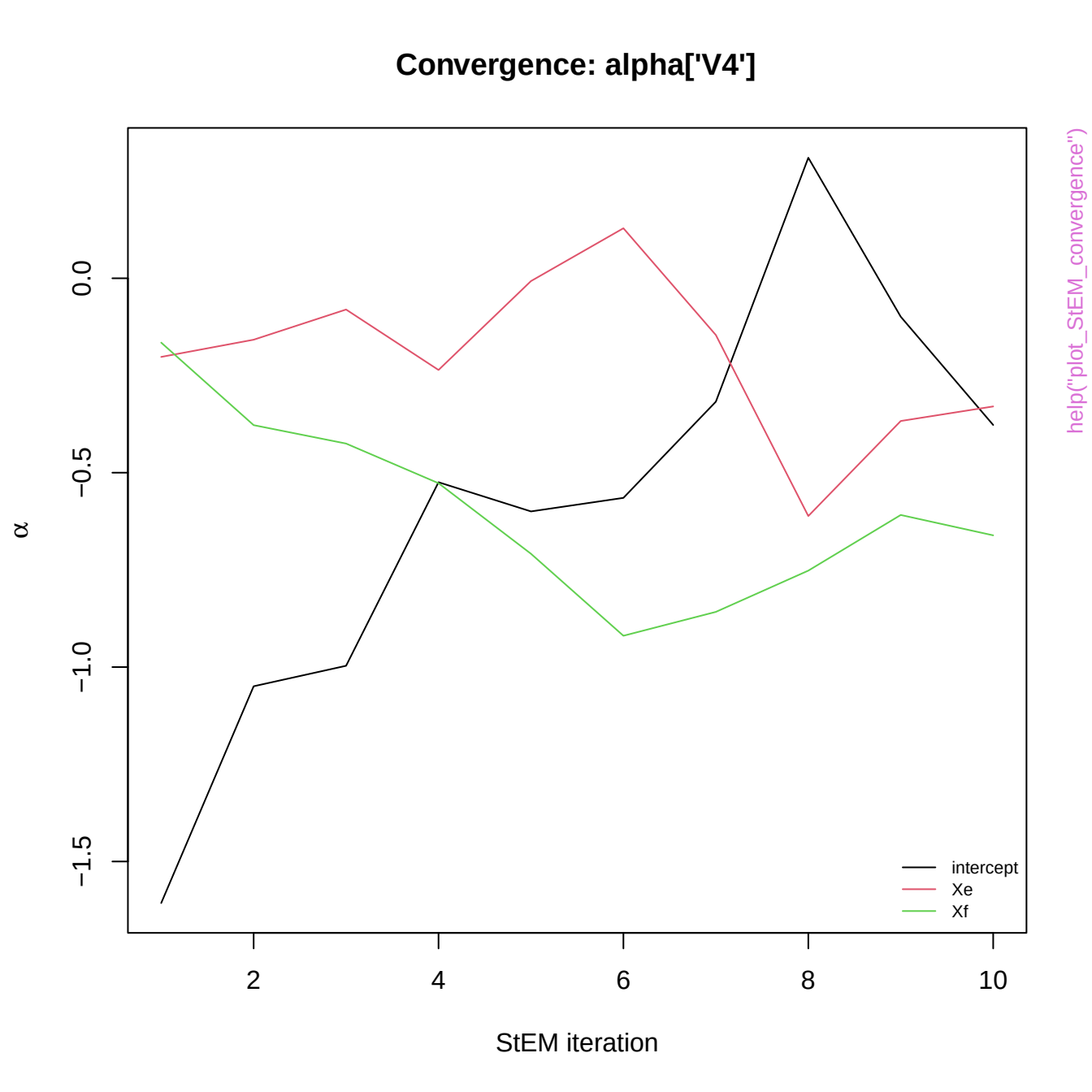

Convergence: gamma (proportion linked)
γ
0.64
0.65
0.66
0.67
0.68
0.69
2
4
6
8
10
StEM iteration
help("plot_StEM_convergence")

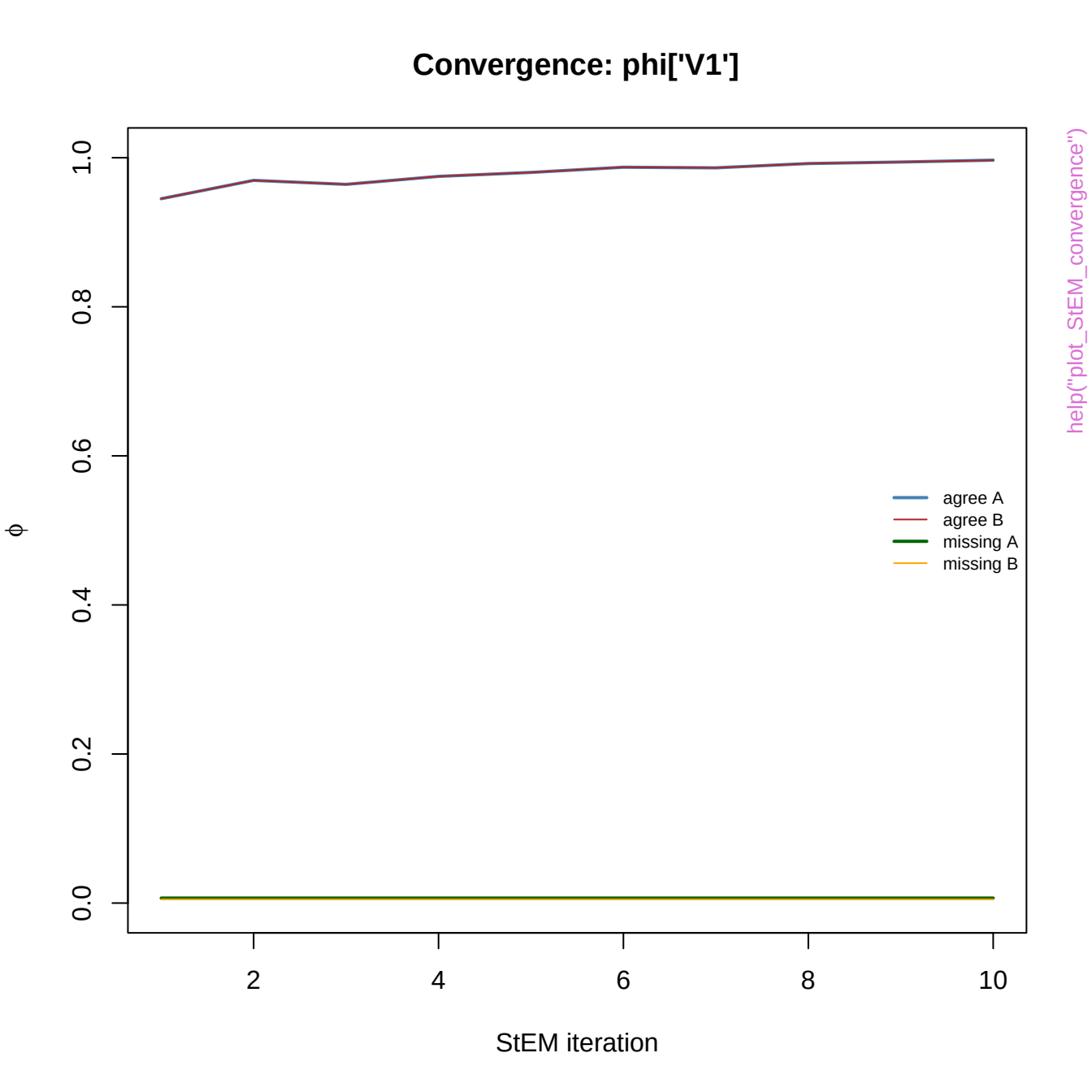

Convergence: eta['V1']
η
0.0
0.1
0.2
0.3
0.4
0.5
2
4
6
8
10
StEM iteration
help("plot_StEM_convergence")

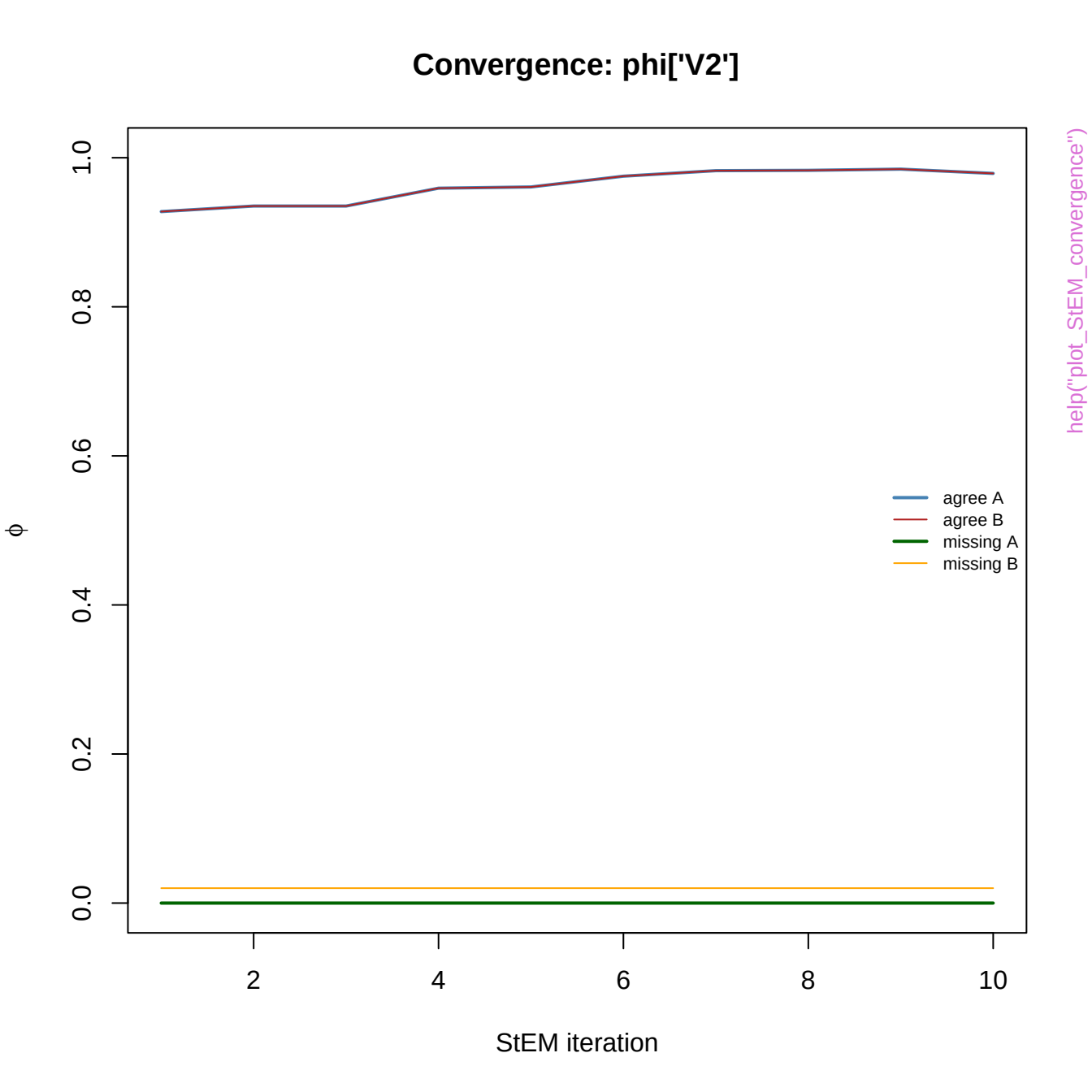

Convergence: eta['V2']
η
0.0
0.1
0.2
0.3
0.4
0.5
2
4
6
8
10
StEM iteration
help("plot_StEM_convergence")

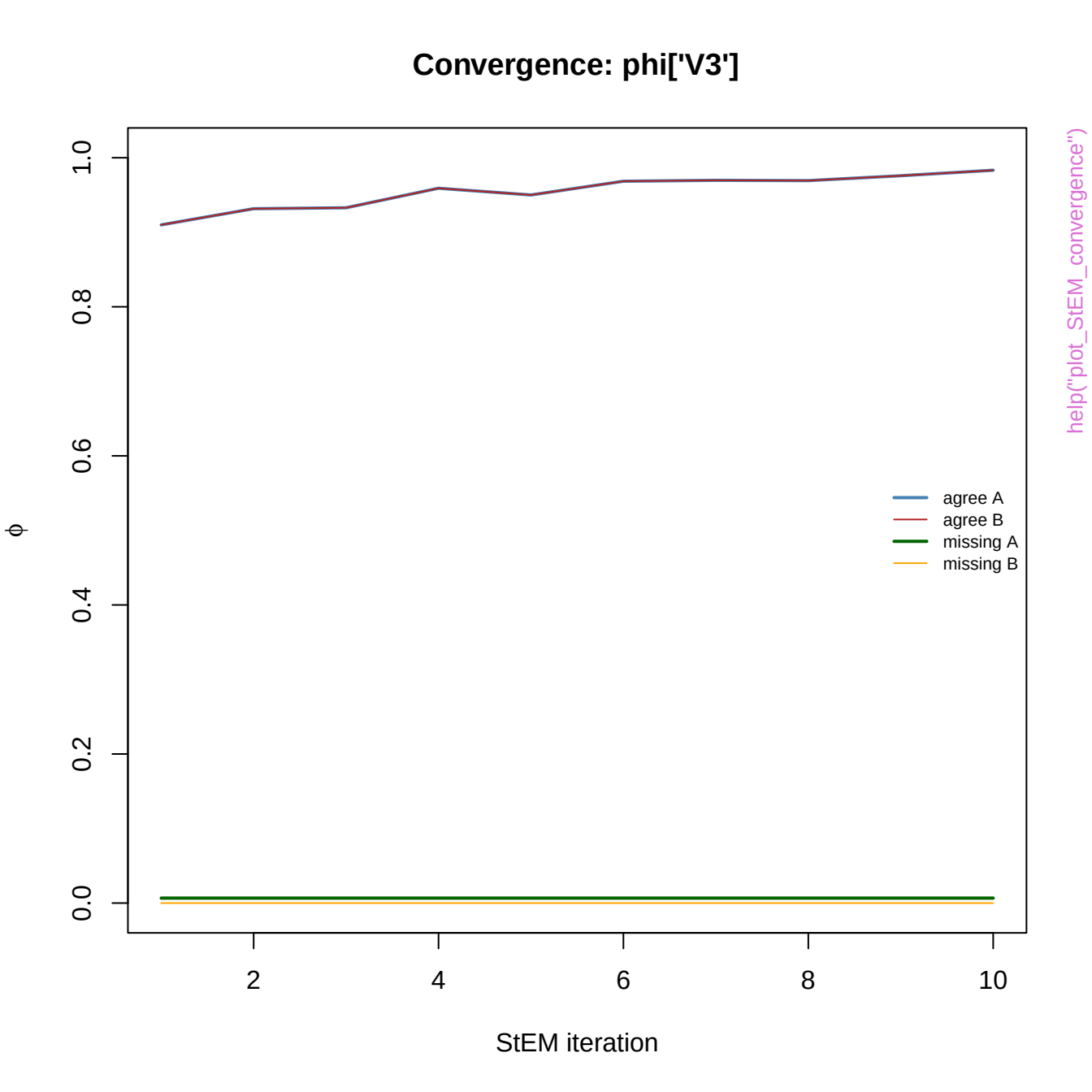

Convergence: eta['V3']
η
0.0
0.1
0.2
0.3
0.4
0.5
2
4
6
8
10
StEM iteration
help("plot_StEM_convergence")

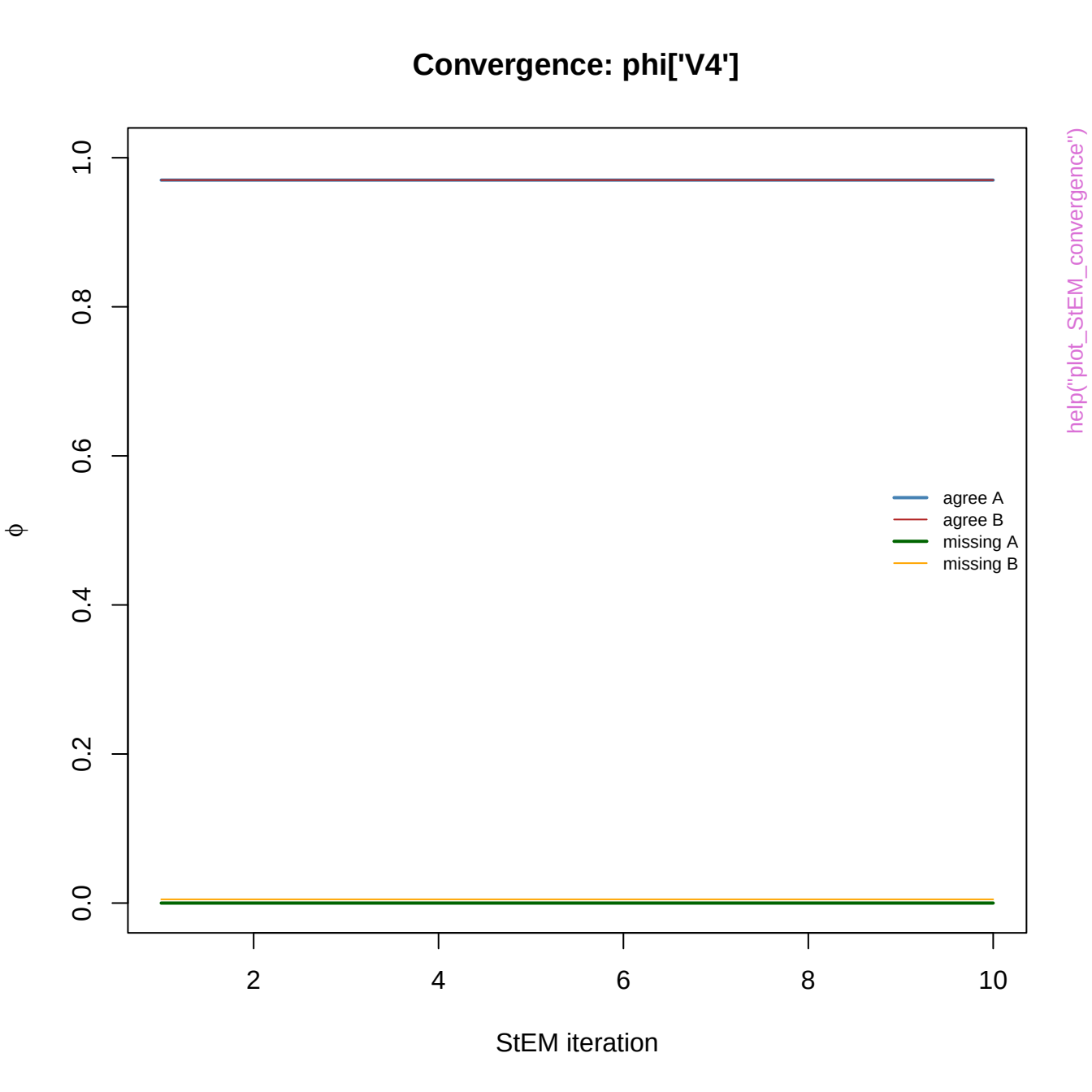

Convergence: eta['V4']
η
0.0
0.1
0.2
0.3
0.4
0.5
2
4
6
8
10
StEM iteration
help("plot_StEM_convergence")

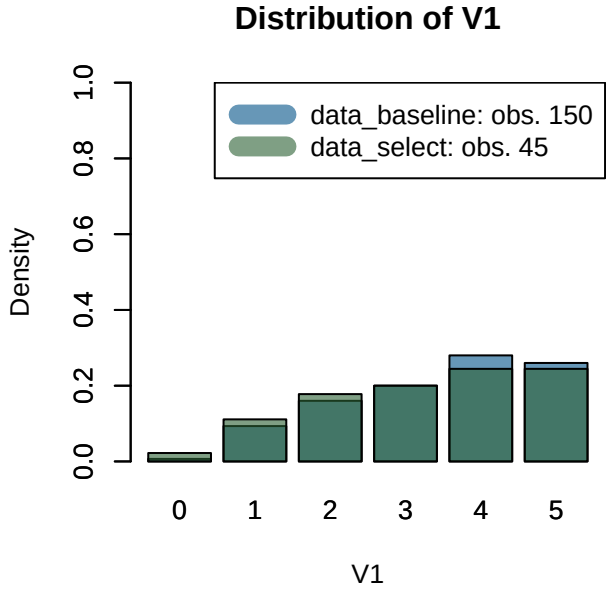

Convergence: alpha['V4']
α
0.0
−0.5
−1.0
−1.5
2
4
6
8
10
StEM iteration
intercept
Xe
Xf
help("plot_StEM_convergence")

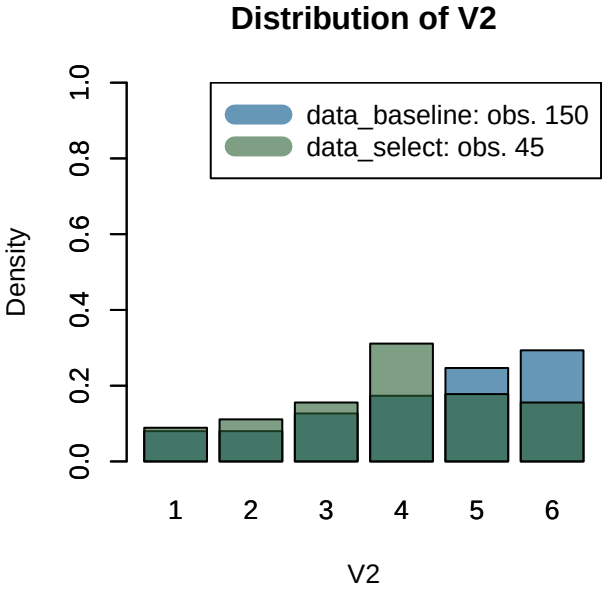

Convergence: phi['V1']
1.0
0.8
0.6
0.4
0.2
0.0
ϕ
agree A
agree B
missing A
missing B
2
4
6
8
10
StEM iteration
help("plot_StEM_convergence")

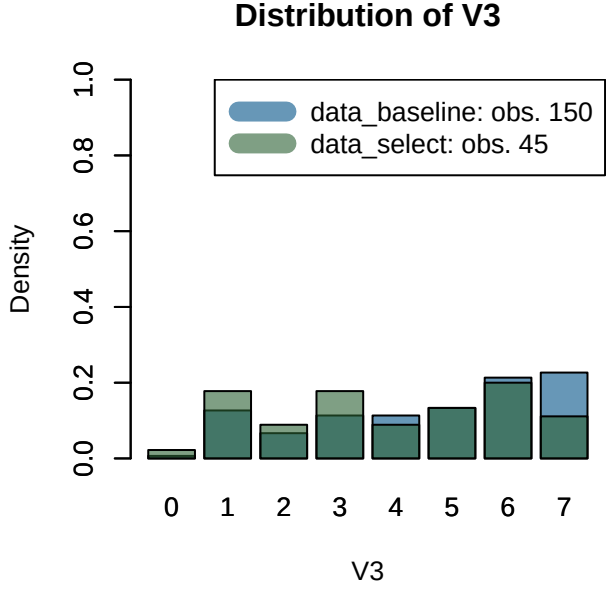

Convergence: phi['V2']
1.0
0.8
0.6
0.4
0.2
0.0
ϕ
agree A
agree B
missing A
missing B
2
4
6
8
10
StEM iteration
help("plot_StEM_convergence")

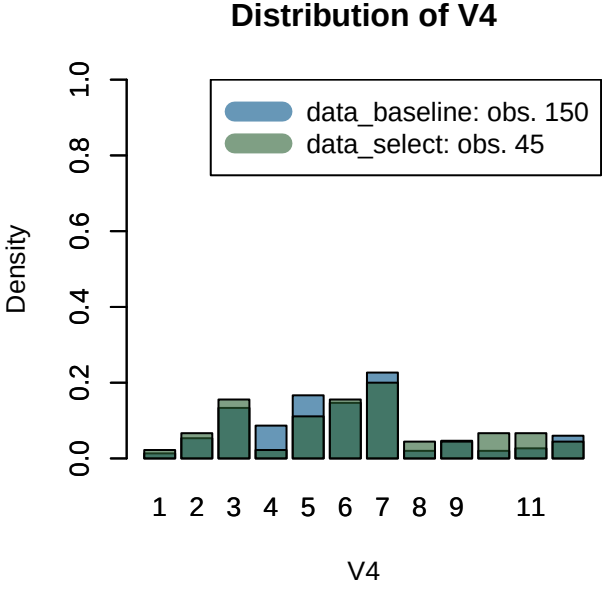

Convergence: phi['V3']
1.0
0.8
0.6
0.4
0.2
0.0
ϕ
agree A
agree B
missing A
missing B
2
4
6
8
10
StEM iteration
help("plot_StEM_convergence")


Convergence: phi['V4']
1.0
0.8
0.6
0.4
0.2
0.0
ϕ
agree A
agree B
missing A
missing B
2
4
6
8
10
StEM iteration
help("plot_StEM_convergence")

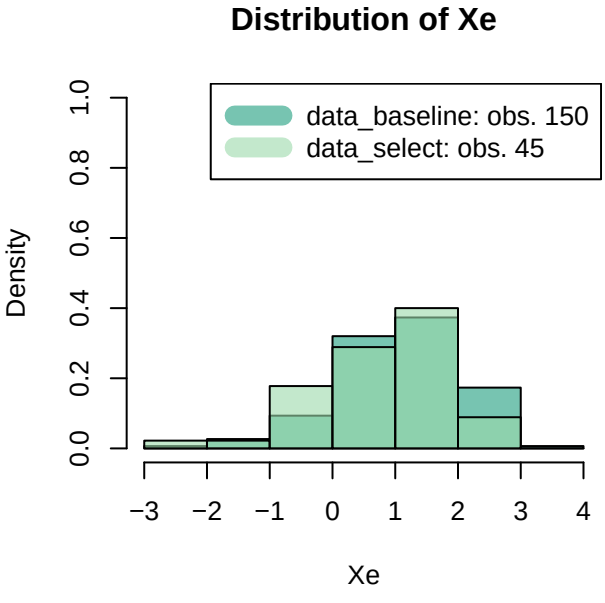


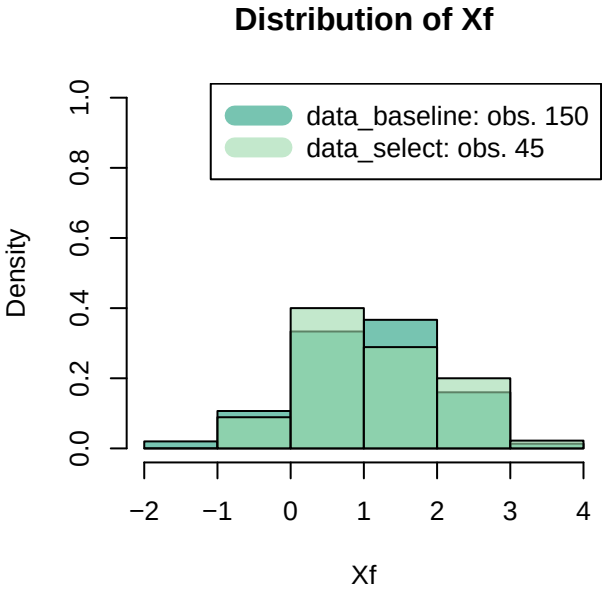


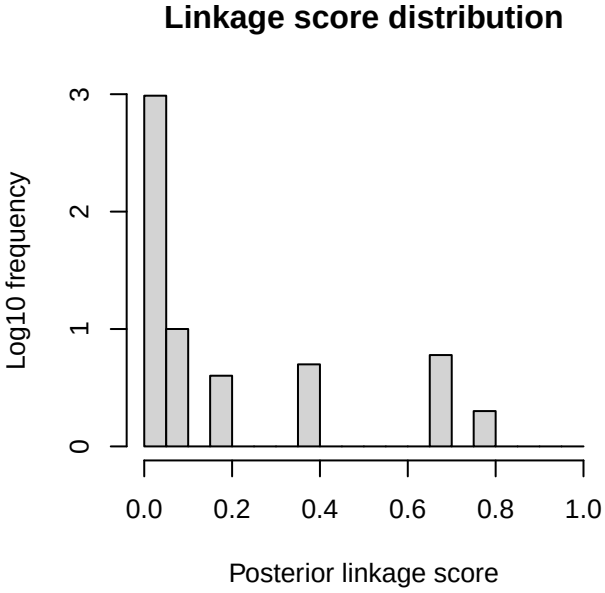




help("plot_distributions")







help("plot_distributions")